AGA CLINICAL PRACTICE GUIDELINE ON PHARMACOLOGICAL INTERVENTIONS FOR MANAGEMENT OF OBESITY

Eduardo Grunvald*[1], Raj Shah*[2], Ruben Hernaez,[3-5*] Apoorva Krishna Chandar[6], Octavia Pickett-Blakely[7], Levi M. Teigen[8], Tasma Harindhanavudhi[9], Shahnaz Sultan[10], Siddharth Singh[11], Perica Davitkov[6,12], on behalf of the AGA Clinical Guidelines Committee

ONLINE SUPPLEMENT

Supplementary Table 1. Problem, Intervention, Comparison, Outcome (PICO) question for this Guideline

Problem	Adults with BMI ≥ 30 kg/m^2, or ≥ 27 kg/m^2 and weight-related complications, who have had an inadequate response to lifestyle interventions
Intervention	FDA-approved medications to treat obesity
Comparison	Lifestyle interventions alone
Outcome	<u>Benefits</u>: ✓ % Total Body Weight loss (%TBWL) ✓ ≥5% Total Body Weight loss ✓ ≥10% Total Body Weight loss <u>Harms</u>: ✓ Treatment discontinuation due to adverse events ✓ Serious Adverse Events <u>Minimal important difference</u> Smallest change in the outcome that would be identified as clinically important was set at -3% TBWL between groups

Abbreviations: Total Body Weight loss (TBWL)

Supplementary Table 2: Serious adverse events from studies, the FDA and contraindications for Semaglutide 2.4 mg

Contraindications [a]	Serious Adverse Reactions FDA [b]	Selected Serious Adverse Reactions original studies' definitions
1. Pregnancy 2. Personal or family history of medullary thyroid carcinoma or in patients with Multiple Endocrine Neoplasia syndrome type 2 2. Known hypersensitivity to semaglutide or any of the excipients in Semaglutide.	1. 131 cases reported (examples: nausea, vomiting, impaired gastric emptying, cholelithiasis, GERD, constipation, thyroid neoplasia, dyspepsia, hypoglycemia, diarrhea, abdominal pain, electrolyte abnormalities/dehydration, LFT abnormality, renal impairment)[c]	1. Wilding 2021: Abdominal pain: I: 3/1306 vs C: 0/655 Constipation: 1/1306 vs 0/655 Diarrhea: 1/1306 vs 0/655 Nausea: 1/1306 vs 0/655 Vomiting: 4/1306 vs 0/655 Pancreatitis: 2/1306 vs 0/655 Vertigo: 3/1306 vs 0/655 Cholelithiasis: 12/1306 vs 1/655 Cholecystitis 4/1306 vs 0/655 Acute MI: 2/1306 vs 1/655 Gastroenteritis: 5/1306 vs 0/655 SI: 1/1306 vs 0/655

a. Product labeling
b. FDA definition: An adverse event when the patient outcome is death, life-threatening, hospitalization, disability of permanent damage, congenital anomaly/birth defect, development of drug dependence or drug abuse (FDA Adverse Event Reporting System Database). See full list on public database.
c. Reported cases are from 2021 to March 31st, 2022. Unknown denominator.

Supplementary Table 3. Evidence Profile for Semaglutide 2.4 mg

Question: Semaglutide plus lifestyle intervention compared to lifestyle intervention alone for management of obesity

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**Semaglutide plus lifestyle intervention**	**lifestyle intervention alone**	**Relative (95% CI)**	**Absolute (95% CI)**		
Total body weight loss (%TBWL) (assessed with: Percentage %)												
8	randomized trials	not serious	not serious[a,b]	not serious	not serious	none	2658	1694	-	**MD 10.76 more (8.73 more to 12.8 more)**	⊕⊕⊕⊕ High	CRITICAL
≥5% TBWL												
6	randomized trials	not serious	not serious[a,c]	not serious	not serious	none	2094/2543 (82.3%)	485/1583 (30.6%)	**RR 2.74 (2.21 to 3.40)**	**533 more per 1,000 (from 371 more to 735 more)**	⊕⊕⊕⊕ High	CRITICAL
≥10% TBWL												
6	randomised trials	not serious	not serious[a,c]	not serious	not serious	none	1651/2543 (64.9%)	195/1583 (12.3%)	**RR 5.25 (3.61 to 7.64)**	**524 more per 1,000 (from 322 more to 818 more)**	⊕⊕⊕⊕ High	CRITICAL
≥15% TBWL												

6	randomised trials	not serious	not serious[a, c]	not serious	not serious	none	1172/2543 (46.1%)	86/1583 (5.4%)	**RR 7.82 (5.19 to 11.76)**	**371 more per 1,000 (from 228 more to 585 more)**	⨁⨁⨁⨁ High	CRITICAL
Weight loss (assessed with kilograms)												
6	randomised trials	not serious	not serious[a, b]	not serious	not serious	none	2544	1584	-	MD **10.81 higher (8.19 higher to 13.43 higher)**	⨁⨁⨁⨁ High	IMPORTANT
Serious Adverse Events (SAE)												
8	randomised trials	not serious	not serious	not serious	serious[d]	none	254/2657 (9.6%)	120/1696 (7.1%)	**RR 1.38 (1.10 to 1.73)**	**27 more per 1,000 (from 7 more to 52 more)**	⨁⨁⨁◯ Moderate	CRITICAL
Treatment discontinuation due to adverse events												
8	randomised trials	not serious	not serious	not serious	not serious[e]	none	170/2657 (6.4%)	52/1696 (3.1%)	**RR 2.10 (1.54 to 2.86)**	**34 more per 1,000 (from 17 more to 57 more)**	⨁⨁⨁⨁ High	CRITICAL

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

Explanations

a. There was inconsistency noted, but it was in the same direction and did not cross the minimally important difference (MID) threshold that we determined a priori at 3%. Thus, we decided to not rate down.

b. The inconsistency can be explained by Davies 2021 including patients with T2DM, while Wilding 2021 does not. Thus, the %TBWL appears to be less in patients who have T2DM.
c. The inconsistency can be explained by Wadden 2021 including intensive behavioral therapy and initial low-calorie diet in both arms. Thus, this minimizes the difference between both arms for this dichotomous outcome
d. Absolute risk crosses threshold of 1%, which was the pre-determined MID threshold.
e. Forced titration expected to be captured in this outcome

Supplementary Table 4. Serious adverse events from studies, the FDA, and contraindications for Liraglutide 3.0 mg

Contraindications [a]	**Serious Adverse Reactions FDA** [b]	**Serious Adverse Reactions original studies' definitions**
1. Pregnancy 2. Patients with a personal or family history of medullary thyroid cancer (MTC) or in patients with Multiple Endocrine Neoplasia syndrome type 2 (MEN 2) 3. Should not be used with other GLP-1 RAs or DPP4 inhibitors 4. In patients treated with insulin or insulin secretagogues (eg sulfonylureas), caution is advised for hypoglycemia and proper dose adjustments may be necessary	1. Pancreatitis: 40 required hospitalizations, and 1 was life-threatening (out of approximately 29,277 who took liraglutide between 2015 and 2018) – 0.1% 2. No life-threatening or fatal breast cancers (between 2015 and 2018) 4. 0 cases of MTC 5. 17 cases of cholelithiasis requiring hospitalization – 0.05% 6. Five cases of renal failure, of which 3 needed hospitalizations, 1 was life-threatening, and 1 death occurred – 0.01% 7. 18 cases of dehydration, of which 16 were hospitalized, 1 life-threatening and 1 caused disability – 0.06% 8. Nausea/vomiting: 36 cases required hospitalization, 1 life-threatening, and 1 death – 0.1% 9. 1 case of hypoglycemia requiring hospitalization 10. 6 cases of serious suicidal behavior (one completed suicide) while the other 5 needed hospitalization or were life-threatening – 0.02%	**Rubino 2022 (STEP 8):** Malignant neoplasms: 3 (including 2 breast cancers) in liraglutide vs. 1 breast cancer in placebo Severe GI adverse events occurred in 3 in liraglutide group vs 3 in placebo group 1 acute pancreatitis in liraglutide vs none in placebo **O'Neil 2018:** 0 acute pancreatitis in liraglutide vs 1 in placebo 3 neoplasms in liraglutide vs 4 in placebo group, but no pancreatic cancer or breast cancer **Lundgren 2021:** 2/49 severe cases of cholelithiasis in liraglutide group vs 0 for placebo 1/49 cases of acute pancreatitis vs 0 in the placebo group **Davies 2015:** 1/422 breast cancer in liraglutide vs 0/212 in placebo group

		Severe hypoglycemia occurred in 3/422 subjects taking liraglutide vs 0/212 subjects taking placebo (in those also on sulfonylureas) **Pi-Sunyer 2015:** 1 participant died in the liraglutide group and 2 in the placebo group. 20/2481 subjects in the liraglutide group had gallstones that were classified as SAE vs 5/1242 in the placebo group 1/2481 acute pancreatitis events with liraglutide that were classified as severe 4/2481 breast cancer in liraglutide vs 1/1242 in placebo (these were advanced node + breast malignancies) 6/2481 participants in liraglutide group had active suicidal ideation (SI) vs 3/1242 in the placebo group. **Garvey 2020:** 3/198 events of severe hypoglycemia with liraglutide vs 2/198 with placebo 6/198 malignant neoplasms in liraglutide vs 2/198 in placebo were considered SAE 0/198 invasive lobular breast cancer with liraglutide vs 1/198 with placebo **Wadden 2013:**

		3/212 had malignant neoplasms (including 2 breast cancers) vs 2/210 in placebo (neither were breast cancer in placebo group) **Astrup 2012:** 1 SAE related to cholelithiasis + acute pancreatitis in liraglutide vs 0 in placebo **Gudbergsen 2021:** 1/80 with GI related SAE in liraglutide (ileus) vs 1/76 GI related SAE in placebo (cholecystitis) **Wadden 2020:** 1 acute cholecystitis, 1 papillary thyroid cancer in liraglutide vs none in placebo

a. Product labeling
b. FDA definition: An adverse event when the patient outcome is death, life-threatening, hospitalization, disability of permanent damage, congenital anomaly/birth defect, development of drug dependence or drug abuse (FDA Adverse Event Reporting System Database)
c. Reported cases are from 2015 to 2018 in the FAERS database. Unknown denominator but one study reports with 29277 users between 2015-2018 who used liraglutide 3.0 mg [ref DOI: 10.1111/dom.14367]

Supplementary Table 5. Evidence Profile for Liraglutide 3.0 mg

Certainty assessment							№ of patients		Effect			
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Liraglutide	Control	Relative (95% CI)	Absolute (95% CI)	Certainty	Importance
Weight loss (follow-up: range 52 weeks to 68 weeks; assessed with: kilograms)												
9	randomized trials	not serious	not serious	not serious	not serious	none	3547	2129	-	MD **5.3 lower (5.9 lower to 4.7 lower)**	⨁⨁⨁⨁ High	IMPORTANT
Percent total body weight loss (follow-up: range 52 weeks to 68 weeks)												
8	randomized trials	not serious	not serious	not serious	not serious	none	3710	2258	-	MD **4.81 higher (4.23 higher to 5.39 higher)**	⨁⨁⨁⨁ High	CRITICAL
Total body weight loss ≥ 5% (follow-up: range 52 weeks to 68 weeks)												
11	randomized trials	not serious	not serious[a]	not serious	not serious	none	2379/3964 (60.0%)	662/2498 (26.5%)	**RR 2.09 (1.80 to 2.42)**	**289 more per 1,000 (from 212 more to 376 more)**	⨁⨁⨁⨁ High	CRITICAL
Total Body Weight Loss ≥ 10% (follow-up: range 52 weeks to 68 weeks)												
11	randomized trials	not serious	not serious	not serious	not serious	none	1208/3953 (30.6%)	259/2497 (10.4%)	**RR 2.67 (2.14 to 3.34)**	**173 more per 1,000 (from 118 more to**	⨁⨁⨁⨁ High	CRITICAL

										243 more)		
Total Body Weight Loss ≥ 15% (follow-up: range 52 weeks to 68 weeks)												
6	randomized trials	not serious	not serious	not serious	not serious	none	439/2958 (14.8%)	77/1704 (4.5%)	**RR 3.04 (2.25 to 4.12)**	**92 more per 1,000 (from 56 more to 141 more)**	⨁⨁⨁⨁ High	IMPORTANT
Serious Adverse Events (follow-up: range 52 weeks to 68 weeks)												
11	randomized trials	not serious	not serious	not serious	serious[b]	none	275/3964 (6.9%)	139/2498 (5.6%)	**RR 1.22 (1.00 to 1.50)**	**12 more per 1,000 (from 0 fewer to 28 more)**	⨁⨁⨁◯ Moderate	CRITICAL
Treatment discontinuations due to adverse effects (follow-up: range 52 weeks to 68 weeks)												
10	randomized trials	not serious	not serious	not serious	not serious	none	373/3914 (9.5%)	98/2448 (4.0%)	**RR 2.31 (1.85 to 2.88)**	**52 more per 1,000 (from 34 more to 75 more)**	⨁⨁⨁⨁ High	CRITICAL

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

Explanations

a. I-squared = 65% and largely influenced by study by Lundgren et al. in which participants had already lost significant body weight during 8-week run-in period, but not rated down as sensitivity analysis by excluding this study did not show a significant change in pooled effect size

b. 95% CI extends from no harms to clinically significant SAE

Supplementary Table 6. Serious adverse events from studies, the FDA and contraindications for Phentermine-topiramate ER

Contraindications	Serious Adverse Reactions FDA [b]	Selected Serious Adverse Reactions original studies' definitions
1. Pregnancy 2. Glaucoma 3. Hyperthyroidism 4. During or within 14 days of taking monoamine oxidase inhibitors 5. Known hypersensitivity or idiosyncrasy to sympathomimetic amines	1. Neurologic (e.g., blurry vision or vision disturbance, seizure, cerebrovascular accident (CVA), paresthesia, dizziness, loss of consciousness, glaucoma, SI, anxiety, change in HR, attention/thinking disturbance): 105 cases reported (~0.6%) [c] 2. General disorders (nephrolithiasis, arrhythmia, vision disturbance, mental disturbance, SI): 79 cases reported (~0.5%) 3. GI (e.g., constipation, abdominal pain, nausea, dry mouth, diarrhea, constipation, change in taste, cholelithiasis, pancreatitis): 51 cases reported (~0.37%) [c] 4. Psychiatric (e.g., anxiety, depression, SI, sleep disorder): 74 cases reported (~ 0.39%) [c] 5. 7 deaths	1. Gadde 2011: Cardiac (coronary artery disease, myocardial infarction (MI), cardio-respiratory arrest, angina pectoris, acute coronary syndrome, atrial fibrillation, tachycardia) Intervention 15 mg/92 mg: 5/994; Placebo: 7/993 Cholelithiasis: I: 1/994 and C: 1/993 Syncope/Headache/Dizziness: I: 1/994 vs C: 3/993 Nephrolithiasis: I: 2/994 vs C: 0/993

a. Product labeling

b. FDA definition: An adverse event when the patient outcome is death, life-threatening, hospitalization, disability of permanent damage, congenital anomaly/birth defect, development of drug dependence or drug abuse (FDA Adverse Event Reporting System Database). See full list on the public database.

c. Reported cases are from 2012 to March 31st, 2022. Unknown denominator but, one study reports with 11,513 users between 2015-2018 that would account for an indirect estimate of 0.3- 1 % SAE [ref DOI: 10.1111/dom.14367]

Supplementary Table 7. Evidence Profile for Phentermine-topiramate ER

Question: Phentermine/Topiramate (15 mg/92 mg) plus lifestyle intervention compared to lifestyle intervention alone for management of obesity

№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Phentermine/Topiramate (15/92 mg) plus lifestyle intervention	lifestyle intervention alone	Relative (95% CI)	Absolute (95% CI)		
Total body weight loss (%TBWL) (assessed with: Percentage %)												
3	randomized trials	not serious	not serious[a]	not serious	not serious	none	1580	1561	-	MD **8.45 % higher (7.89 higher to 9.01 higher)**	⊕⊕⊕⊕ High	CRITICAL
≥5% TBWL												
3	randomized trials	not serious	not serious[a]	not serious	not serious	none	1068/1580 (67.6%)	303/1561 (19.4%)	**RR 3.48 (3.13 to 3.87)**	**481 more per 1,000 (from 413 more to 557 more)**	⊕⊕⊕⊕ High	CRITICAL
≥10% TBWL												
3	randomized trials	not serious	not serious[a]	not serious	not serious	none	730/1580 (46.2%)	114/1561 (7.3%)	**RR 6.33 (5.26 to 7.61)**	**389 more per 1,000 (from 311 more to 483 more)**	⊕⊕⊕⊕ High	CRITICAL
≥15% TBWL												
1	randomized trials	not serious	not serious	not serious	serious[b]	none	161/511 (31.5%)	17/513 (3.3%)	**RR 9.51 (5.86**	**282 more per**	⊕⊕⊕◯	CRITICAL

									to 15.44)	**1,000 (from 161 more to 479 more)**	Moderate	
Serious Adverse Events (SAE)												
3	randomized trials	not serious	not serious	not serious	serious[b,c]	none	67/1580 (4.2%)	55/1561 (3.5%)	**RR 1.20 (0.85 to 1.70)**	**7 more per 1,000 (from 5 fewer to 25 more)**	⊕⊕⊕◯ Moderate	CRITICAL
Treatment discontinuation due to adverse events												
3	randomized trials	not serious	not serious	not serious	not serious	none	275/1580 (17.4%)	132/1561 (8.5%)	**RR 2.08 (1.71 to 2.52)**	**91 more per 1,000 (from 60 more to 129 more)**	⊕⊕⊕⊕ High	CRITICAL

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

Explanations

a. Allison 2012; BMI ≥35
b. Less than 300 events
c. crosses unity

Supplementary Table 8. SAE FAERS for Bupropion-naltrexone ER

Contraindications [a]	Serious Adverse Reactions FDA [b]	Serious Adverse Reactions original studies' definitions [d]
1. Chronic opioid therapy 2. Need for short-term opioid analgesics 3. Uncontrolled hypertension 4. Seizure disorder, history of seizures, or risk of seizures 5. Use during or within 14 days following MAO inhibitor 6. Pregnancy 7. While not contraindication, check drug-drug interactions	1. Neurologic (e.g., dizziness, seizure, CVA, headache): 93 cases reported (~0.1%) [c] 2. GI (e.g., nausea, vomiting, diarrhea, constipation, cholecystitis): 103 cases reported (~0.1%) [c] 3. Psychiatric (e.g., anxiety, depression, SI): 75 cases reported (> 0.1%) [c]	**1. Apovian 2013** 1/992 (0.1%) passive suicidal ideation, 1/992 (0.1%) seizure 2. **Wadden 2010** 2/584 (0.3%): cholecystitis 3. **Nissen 2016** CV events: all death, nonfatal stroke, MI: 40/4455 (0.9%) vs 62/4450 (1.4%) in placebo 4. **Greenway 2010** 2/573 (0.3%): cardiac (1 heart failure, 1 MI) 0/573 stroke 5. **Internal SR MA** seizure - 0.1% vs 0% in placebo, SI in 1/ 3,239 (0.03%) vs 3/1,515 (0.20%) in placebo

a. Product labeling
b. FDA definition: An adverse event when the patient outcome is death, life-threatening, hospitalization, disability of permanent damage, congenital anomaly/birth defect, development of drug dependence or drug abuse (FDA Adverse Event Reporting System Database)
c. Reported cases are from 2014 to Dec 31st, 2021. Unknown denominator but, one study reports with 42 138 users between 2015-2018 that would account for 0.5-1 % all SAE [ref DOI: 10.1111/dom.14367]
d. Nissen: CV events, including all death, nonfatal stroke, or nonfatal MI infarction and hospitalization, Greenway: by good clinical practice (same as FDA)

Supplementary Table 9. Evidence Profile for Naltrexone-bupropion ER

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	NB32	No treatment	Relative (95% CI)	Absolute (95% CI)		
Weight loss in kg												
5	randomized trials	not serious[a,b]	not serious[c]	not serious	serious[d,e]	none	6772	5887	-	MD **3.01 kg fewer (3.39 fewer to 2.62 fewer)**	⨁⨁⨁◯ Moderate	IMPORTANT
%TBWL												
5	randomized trials	not serious[a,b]	not serious[c]	not serious	serious[d,e]	none	6772	5887	-	MD **3.01 % lower (3.54 lower to 2.47 lower)**	⨁⨁⨁◯ Moderate	CRITICAL
TBWL >5%												
4	randomized trials	not serious	not serious[f]	not serious	not serious	none	949/2317 (41.0%)	247/1437 (17.2%)	**RR 2.18 (1.41 to 3.37)**	**203 more per 1,000 (from 70 more to 407 more)**	⨁⨁⨁⨁ High	CRITICAL
TBWL >10%												
4	randomized trials	not serious	not serious[f]	not serious	not serious	none	559/2317 (24.1%)	105/1437 (7.3%)	**RR 3.04 (1.80 to 5.14)**	**151 more per 1,000 (from 58 more to 303 more)**	⨁⨁⨁⨁ High	CRITICAL

TBWL >15%												
3	randomized trials	not serious	not serious[f]	not serious	not serious	none	279/2052 (13.6%)	39/1278 (3.1%)	**RR 3.88 (2.13 to 7.08)**	**88 more per 1,000 (from 34 more to 186 more)**	⨁⨁⨁⨁ High	IMPORTANT
Treatment discontinuation due to AE												
5	randomized trials	not serious	not serious[g]	not serious	not serious	none	1798/6947 (25.9%)	545/5892 (9.2%)	**RR 2.39 (1.69 to 3.37)**	**129 more per 1,000 (from 64 more to 219 more)**	⨁⨁⨁⨁ High	CRITICAL
Serious adverse events												
5	randomized trials	not serious	not serious	not serious	serious[h]	none	67/6947 (1.0%)	78/5892 (1.3%)	**RR 0.74 (0.53 to 1.03)**	**3 fewer per 1,000 (from 6 fewer to 0 fewer)**	⨁⨁⨁◯ Moderate	CRITICAL

a. Apovian 2013, had imputation in the analysis for week 56. After re-randomization, patients receiving a higher dose of Naltrexone were excluded from the final analysis, and participants receiving FDA approved dose were counted twice "double-weighted". These were not many participants, and the panel did not estimate they would impact the pool estimate.
b. To deal with attrition bias most studies used intention to treat (ITT) and total original number was used as the last observation carried forward. However, Hollander et al report only on modified ITT analysis and excluded subjects that discontinued the drug within first 4 weeks, the majority due to side effects. We explored this in a sensitivity analysis and there was no large difference between the results. Thus, we decided not to rate down for risk of bias (RoB)
c. Although I^2 was high, all the studies were showing benefit, so we did not rate down for inconsistency
d. Minimal important difference (MID) or clinically important threshold below which there is no clear benefit of the intervention in discussion with the guideline panel and technical review team was determined to be 3kg (or ~3%)
e. We noted serious imprecision as the lower confidence limit crosses the minimal important difference (MID) for benefit.
f. Although I^2 was high, all studies showed clear benefits. Furthermore, the Wadden study that was inconsistent with the results from the other studies was conducted with high intensity behavior modification that may have lead to a ceiling effect. However, we performed a sensitivity analysis to explore and the results were not significantly different
g. Although I^2 was high, all the studies were showing harm, so we did not rate down for inconsistency
h. Low event rate and CI crossed the line of no effect

Supplementary Table 10. Serious adverse events from studies, the FDA and contraindications for orlistat.

Contraindications[a]	**FAERS data on Orlistat from 2015 to 2018** [b]	**Selected Serious Adverse Reactions original studies' definitions**
1. Pregnancy 2. Patients with chronic malabsorption syndrome 3. Patients with cholestasis 4. Patients are strongly encouraged to take a multivitamin supplement that contains fat-soluble vitamins to ensure adequate nutrition because Orlistat has been shown to reduce the absorption of fat-soluble vitamins and beta-carotene	1. Liver failure: 12 cases outside the USwith Orlistat 120mg and 1 case in the US with Orlistat 60 mg reported between April 1999 and August 2009 out of an estimated 40 million people worldwide who have used either of these medications 2. There were 4 cases of pancreatic cancer, of which 2 died 3. 4 cases of renal failure reported among 1341 patients (only in 2017 in FAERS) [c]	Davidson 1999: no major SAE in either Orlistat or placebo groups. Broom 2002 (one of the larger studies with 265 subjects in orlistat and 266 in placebo – none of the SAE were thought to be attributable to Orlistat. Sjöström 1998: SAE were reported by 25 out of 345 participants on Orlistat vs 24 out of 343 patients on placebo in year 1, of which only one SAE was judged by the investigators to be related to treatment. Similarly, 2 SAE were judged by the investigators to be possibly related to treatment during year 2. Torgerson (XENDOS) 2004: Over the 4-year study period, a similar proportion of Orlistat and placebo-treated subjects had at least one SAE (15 vs. 13%) Lindgärde 2000: A total of 19 participants out of 190 in the orlistat group and 5 out of 186 in the placebo group experienced SAE, none of which were deemed by the investigators to have a causal relationship with the study medication.

a. Product labeling
b. FDA definition: An adverse event when the patient outcome is death, life-threatening, hospitalization, disability of permanent damage, congenital anomaly/birth defect, development of drug dependence or drug abuse (FDA Adverse Event Reporting System Database)

Supplementary Table 11. Evidence Profile for Orlistat

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Orlistat	Controls	Relative (95% CI)	Absolute (95% CI)		
Weight loss (follow-up: range 48 weeks to 4 years; assessed with: kilograms)												
23 (24 cohorts)	randomized trials	not serious[a]	not serious	not serious	serious[b]	none	4166	3601	-	MD **2.81 lower (3.45 lower to 2.17 lower)**	⊕⊕⊕◯ Moderate	IMPORTANT
Percent total body weight loss (follow-up: range 48 weeks to 4 years)												
15 (16 cohorts)	randomized trials	not serious[a]	not serious[c]	not serious	serious[b]	none	3514	3004	-	MD **2.78 higher (2.36 higher to 3.2 higher)**	⊕⊕⊕◯ Moderate	CRITICAL
Total body weight loss ≥5% (follow-up: range 48 weeks to 4 years)												
18	randomized trials	not serious	not serious[d]	not serious	not serious	none	3303/5638 (58.6%)	1824/5201 (35.1%)	**RR 1.71 (1.55 to 1.88)**	**249 more per 1,000 (from 193 more to 309 more)**	⊕⊕⊕⊕ High	CRITICAL
Total body weight loss ≥10% (follow-up: range 48 weeks to 4 years)												
15	randomized trials	not serious	not serious[e]	not serious	not serious	none	1577/5083 (31.0%)	715/4652 (15.4%)	**RR 1.94 (1.70 to 2.22)**	**144 more per 1,000 (from 108**	⊕⊕⊕⊕ High	CRITICAL

										more to 188 more)		
Severe adverse events (follow-up: range 48 weeks to 4 years)												
11	randomized trials	not serious	not serious	not serious	serious[f]	none	362/3542 (10.2%)	329/3545 (9.3%)	**RR 1.04 (0.81 to 1.33)**	**4 more per 1,000 (from 18 fewer to 31 more)**	⊕⊕⊕◯ Moderate	CRITICAL
Treatment discontinuations due to adverse events (follow-up: range 48 weeks to 4 years)												
20	randomized trials	not serious	not serious[g]	not serious	not serious	none	339/4252 (8.0%)	195/3804 (5.1%)	**RR 1.51 (1.22 to 1.89)**	**26 more per 1,000 (from 11 more to 46 more)**	⊕⊕⊕⊕ High	CRITICAL

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

Explanations

a. Although most studies did not explain the method of randomization or how allocation was concealed, we did not think it would introduce serious RoB as most studies were double-blinded and although there was attrition seen in several studies, it was usually not disproportionate and LOCF analysis method used for ITT analysis. Thus we did not rate down for RoB
b. 95% CI includes minimal clinically important difference
c. There was moderate heterogeneity (I^2 = 39%), but not rated down as the inconsistency was largely driven by the magnitude and not the direction of effects
d. There was substantial heterogeneity (I^2 = 73%), but not rated down as the inconsistency was largely driven by the magnitude and not the direction of effects
e. There was moderate heterogeneity (I^2 = 46%), but not rated down as the inconsistency was mostly due to the magnitude and not the direction of effects
f. 95% CI is wide and includes 1
g. There was moderate heterogeneity present (I^2 = 30%), but not rated down as the inconsistency was largely driven by the magnitude and not the direction of effects

Supplementary Table 12. Serious adverse events from studies, the FDA and contraindications for phentermine.

Contraindications [a]	Serious Adverse Reactions FDA [b, c]	Serious Adverse Reactions original studies' definitions [e]
1. History of cardiovascular disease (arrhythmias, heart failure, CAD, stroke, uncontrolled hypertension) 2. Hyperthyroidism 3. Glaucoma 4. Agitated states 5. History of drug abuse 6. Use during or within 14 days following MAO inhibitor 7. Pregnancy and/ or breast-feeding	1. Cardiac disorders (e.g., valve disease, MI, arrhythmias CHF, palpitations, tachycardia): 2,554 cases reported (~ 0.05% incidence) 2. Primary pulmonary hypertension - 672 cases reported [d] (~ 0.0008%) 3. CNS effects (e.g., delirium, mania, psychosis, insomnia, irritability, and anxiety): 1,434 cases (~0.01%) 4. Neurologic (dizziness, headache, amnesia, syncope CVA): 1,394 cases reported (~0.01%)	**Aronne 2013** 1/109 (1%): chest pain **Kim 2006** 3/28 (10%): severe headache and nausea 1/28 (3%): dry mouth **Tsai 2012** 3/ 23 (13%): clinically significant BP elevation 1/ 23 (4%): HR elevation

a. Product labeling

b. FDA definition: An adverse event when the patient outcome is death, life-threatening, hospitalization, disability of permanent damage, congenital anomaly/birth defect, development of drug dependence or drug abuse (FDA Adverse Event Reporting System Database)

c. All SAE reported cases are from 1975 to 2021. Unknown denominator but is widely prescribed AOM in the United States, with 116,435 users between 2015-2018 [doi.org/10.1111/dom.14367]

d. Appears to be associated with more extended use. Pulmonary hypertension and valvulopathies were noted in patients using phentermine in combination with fenfluramine.

e. No definitions were provided for SAE in any of the studies

Supplementary Table 13. Evidence Profile for Phentermine

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other	Phentermine/ diet/ exercise	SOC (diet/exercise)	Relative (95% CI)	Absolute (95% CI)		
Weight loss (kg)												
7	randomized trials	not serious[a]	serious[b,c]	serious[d]	not serious	none	353	343	-	MD **4.74 kg more (5.73 more to 3.75 more)**	⊕⊕◯◯ Low	IMPORTANT
%TBWL												
3	randomized trials	not serious[a]	not serious	serious[d]	serious[b,e]	none	205	202	-	MD **3.63 % more (4.29 more to 2.97 more)**	⊕⊕◯◯ Low	CRITICAL
> 5% TBWL												
5	randomized trials	not serious	not serious	serious[d]	not serious	none	173/322 (53.7%)	40/313 (12.8%)	**RR 4.12 (3.04 to 5.59)**	**399 more per 1,000 (from 261 more to 587 more)**	⊕⊕⊕◯ Moderate	CRITICAL
>10% TBWL												
5	randomized trials	not serious	serious[f]	serious[d]	not serious[g]	none	81/322 (25.2%)	15/313 (4.8%)	**RR 5.10 (3.02 to 8.61)**	**196 more per 1,000 (from 97**	⊕⊕◯◯ Low	CRITICAL

										more to 365 more)		
Treatment discontinuation due to side effects												
7	randomized trials	not serious[h]	not serious	serious[d]	not serious	none	257/1274 (20.2%)	76/730 (10.4%)	**RR 1.73 (1.36 to 2.19)**	**76 more per 1,000 (from 37 more to 124 more)**	⨁⨁⨁◯ Moderate	CRITICAL
Severe Adverse Events												
5	randomized trials	not serious[h]	not serious	serious[d]	serious[i]	none	11/280 (3.9%)	3/276 (1.1%)	**RR 2.44 (0.60 to 10.03)**	**16 more per 1,000 (from 4 fewer to 98 more)**	⨁⨁◯◯ Low	CRITICAL

a. Although there was a concern for attrition bias, in almost all the studies that have attrition, it was very similar between the 2 groups. Thus, we did not rate down for RoB

b. Minimal important difference (MID) or clinically important threshold below which there is no clear benefit of the intervention in discussion with the guideline panel and technical review team was determined to be 3 kg (or ~3%)

c. There is a serious inconsistency between the studies because some studies are showing clear benefits with both upper and lower confidence intervals being above the MID, while other studies failed to show clear clinical benefit as the point estimate and the lower confidence limit are below the MID

d. Serious indirectness in the intervention duration. Our PICO question is weight loss treatment for a long-term therapy that most studies define as 1 year. However, the available data is mostly 3-6 months.

e. Confidence interval is crossing the MID, the lower CI is showing clear benefits while the upper failed to show clear clinical benefit as higher confidence limit is below the MID

f. Serious inconsistency possibly due to different intervention duration and follow-up time

g. Concern for serious imprecision due to wide CI, but because this was most likely related to the inconsistency already accounted for, it was decided not to rate down for imprecision

h. Although some smaller and older studies did not blind, overall, there was no concern for serious RoB across the pool of evidence

i. Low event number

Supplementary Table 14. Evidence Profile for Diethylpropion

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Diethylpropion + SOC (diet and exercise)	SOC (diet and exercise)	Relative (95% CI)	Absolute (95% CI)		
Weight loss (kg)												
6	randomized trials	serious[a]	serious[b,c]	serious[d]	not serious	none	209	199	-	MD **4.74 kg more (6.4 more to 3.08 more)**	⊕◯◯◯ Very low	CRITICAL
% TBWL												
4	randomized trials	serious[a]	not serious	serious[d]	serious[e]	none	119	108	-	MD **5.36 % more (7.23 more to 3.5 more)**	⊕◯◯◯ Very low	CRITICAL
>5% TBWL												
3	randomized trials	not serious	not serious	serious[d]	serious[f]	none	109/143 (76.2%)	26/139 (18.7%)	**RR 3.51 (1.50 to 8.18)**	**469 more per 1,000 (from 94 more to 1,000 more)**	⊕⊕◯◯ Low	CRITICAL
>10% TBWL												
3	randomized trials	not serious	not serious	serious[d]	serious[f]	none	71/143 (49.7%)	3/139 (2.2%)	**RR 14.48 (5.13**	**291 more per 1,000**	⊕⊕◯◯ Low	CRITICAL

									to 40.80)	**(from 89 more to 859 more)**		
Treatment discontinuation due to side effects												
5	randomized trials	not serious	not serious	serious[d]	serious[f]	none	10/187 (5.3%)	6/180 (3.3%)	**RR 1.37 (0.51 to 3.66)**	**12 more per 1,000 (from 16 fewer to 89 more)**	⊕⊕◯◯ Low	CRITICAL

a. Majority of the older studies did not perform ITT analyses and only reported the benefit data from the subjects who completed the study, which probably introduced a RoB and overestimated the intervention's effect. In addition, it was unclear how the randomization and allocation were conducted in most of the studies

b. Minimal important difference (MID) or clinically important threshold below which there is no clear benefit of the intervention in discussion with the guideline panel and technical review team was determined to be 3 kg (or ~3%)

c. There is a serious inconsistency between the studies because some studies are showing clear benefits with both upper and lower confidence intervals being above the MID, while other studies failed to show clear clinical benefit as the point estimate and the lower confidence limit are below the MID

d. Serious indirectness in the intervention duration. Our PICO question is weight loss treatment for a long-term therapy that most studies define as 1 year. However, the available data is mostly 3-6 months.

e. Small sample size (<400 total participants)

f. Number of events is low (<200 events)

Supplementary Table 15. Evidence Profile for Gelesis100

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Cellulose and citric acid hydrogel plus lifestyle intervention	lifestyle intervention alone	Relative (95% CI)	Absolute (95% CI)		
Total body weight loss (%) (follow-up: 24 weeks; assessed with: Percentage))												
1	randomized trials	not serious	not serious	not serious	very serious[a,b]	none	223	213	-	MD **2.02 % more (0.96 more to 3.08 more)**	⊕⊕◯◯ Low	
At least 5% TBWL (follow-up: 24 weeks)												
1	randomized trials	not serious	not serious	not serious	serious[c]	none	130/223 (58.3%)	90/213 (42.3%)	**RR 1.38 (1.14 to 1.67)**	**161 more per 1,000 (from 59 more to 283 more)**	⊕⊕⊕◯ Moderate	
At least 10% (follow-up: 24 weeks)												
1	randomised trials	not serious	not serious	not serious	serious[c]	none	61/223 (27.4%)	32/213 (15.0%)	**RR 1.82 (1.24 to 2.67)**	**123 more per 1,000 (from 36 more to 251 more)**	⊕⊕⊕◯ Moderate	
Treatment discontinuation due to adverse events (follow-up: 24 weeks)												

1	randomized trials	not serious	not serious	not serious	very serious[c,d]	none	8/223 (3.6%)	7/213 (3.3%)	**RR 1.09 (0.40 to 2.96)**	**3 more per 1,000 (from 20 fewer to 64 more)**	⊕⊕◯◯ Low	
Serious Adverse Events												
1	randomized trials	not serious	not serious	not serious	very serious[b,c,d]	none	0/223 (0.0%)	1/213 (0.5%)	**RR 0.32 (0.01 to 7.77)**	**3 fewer per 1,000 (from 5 fewer to 32 more)**	⊕⊕◯◯ Low	

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

Explanations

a. Does not meet minimally important threshold of 3%
b. Wide confidence interval
c. Low event rate (<300 events)
d. Crosses unity

Supplementary Figure 1. Search strategy for the Guideline

PICO 1-6

(("Obesity"[Mesh] OR "Weight Loss"[Mesh] OR "Overweight"[Mesh] OR obes*[tw] OR "body mass ind*"[tw] OR adiposity[tw] OR overweight[tw] OR "over weight"[tw] OR "overload syndrome*"[tw] OR "over eat*"[tw] OR overfeed*[tw] OR "over feed*"[tw] OR overfed[tw] OR "over fed"[tw] OR "weight cycling"[tw] OR "skinfold thickness"[tw] OR antiobesity[tw] OR "anti-obesity"[tw] OR obesitas[tw] OR bodyweight[tw] OR "body weight"[tw] OR adipositas[tw] OR BMI[tw]) AND (alli[tw] OR orlipastat[tw] OR orlistat[tw] OR "ro 18 0647"[tw] OR "ro 180647"[tw] OR ro180647[tw] OR tetrahydrolipstatin[tw] OR Xenical[tw] OR (phentermine[tw] AND topiramate[tw]) OR "phentermine topiramate"[tw] OR phenterminetopiramate[tw] OR qnexa[tw] OR qsiva[tw] OR Qsymia[tw] OR topiramatephentermine[tw] OR "phentermine-topiramate"[tw] OR (amfebutamone[tw] AND naltrexone[tw]) OR (bupropion[tw] AND naltrexone[tw]) OR Contrave[tw] OR mysimba[tw] OR "nb 32"[tw] OR nb32[tw] OR "bupropion-naltrexone"[tw] OR liraglutide[tw] OR "nn 2211"[tw] OR nn2211[tw] OR "nnc 90 1170"[tw] OR "nnc90 1170"[tw] OR Saxenda[tw] OR Victoza[tw] OR semaglutide[tw] OR nn9535[tw] OR "nn 9535"[tw] OR ozempic[tw] OR rybelsus[tw] OR "Orlistat"[Mesh] OR "Liraglutide"[Mesh] OR "semaglutide" [Supplementary Concept] OR "bupropion hydrochloride, naltrexone hydrochoride drug combination" [Supplementary Concept])) NOT (("Child"[Mesh] OR "Infant"[Mesh] OR "Adolescent"[Mesh]) NOT "Adult"[Mesh]) AND ("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tw] OR randomised[tw] OR placebo[tw] OR "drug therapy"[sh] OR randomly[tw] OR trial[tw] OR groups[tw]) NOT ("Animals"[sh] NOT "Humans"[sh]) AND ("2021/01/01"[Date - Publication] : "3000"[Date - Publication])

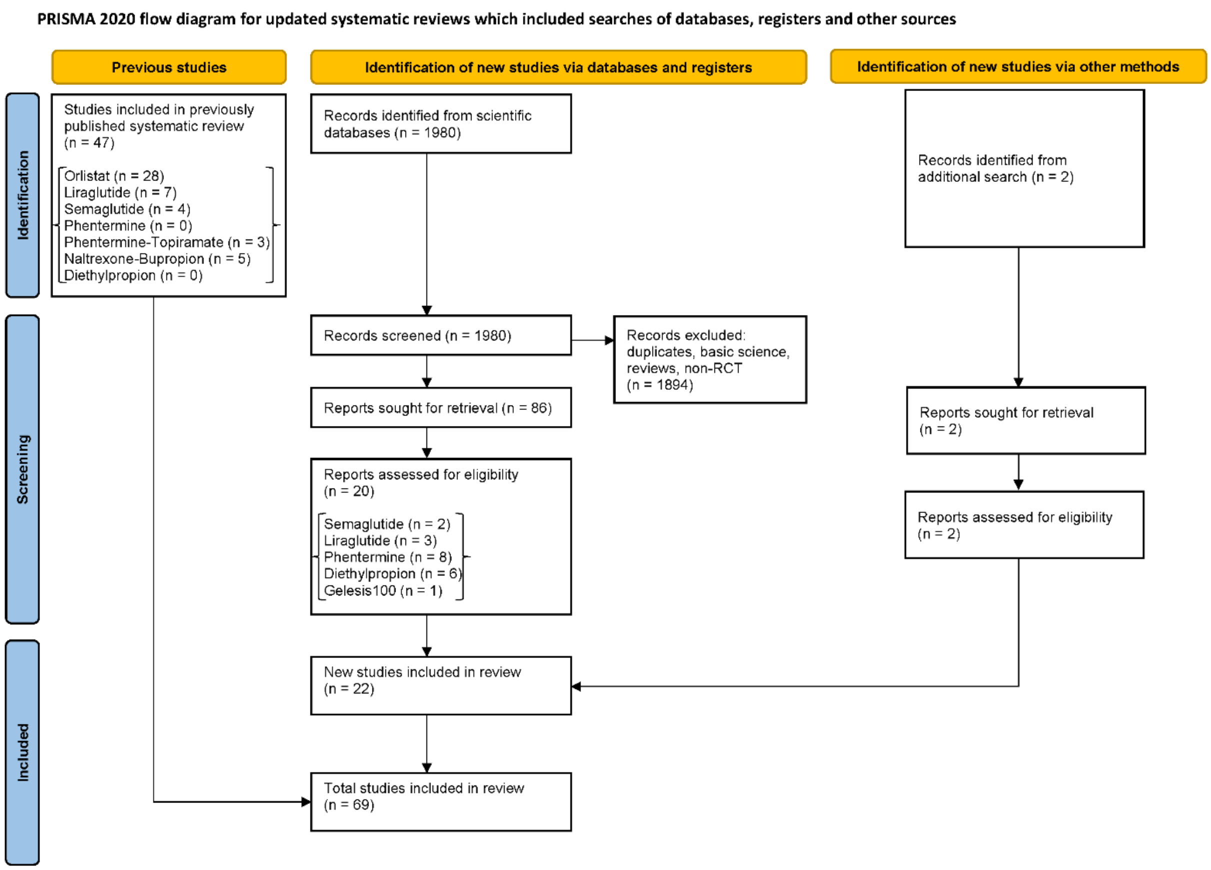
PICO 7 and 8

("Obesity"[Mesh] OR "Weight Loss"[Mesh] OR "Overweight"[Mesh] OR obes*[tw] OR "body mass ind*"[tw] OR adiposity[tw] OR overweight[tw] OR "over weight"[tw] OR "overload syndrome*"[tw] OR "over eat*"[tw] OR overfeed*[tw] OR "over feed*"[tw] OR overfed[tw] OR "over fed"[tw] OR "weight cycling"[tw] OR "skinfold thickness"[tw] OR antiobesity[tw] OR "anti-obesity"[tw] OR obesitas[tw] OR bodyweight[tw] OR "body weight"[tw] OR adipositas[tw] OR BMI[tw]) AND ("Phentermine"[Mesh] OR phentermine[tw] OR 0k2i505otv[tw] OR "1197-21-3 (hcl)"[tw] OR "122-09-8 (phentermine)"[tw] OR "adipex p"[tw] OR "adipex-p"[tw] OR adipexp[tw] OR c045tql4wp[tw] OR duromine[tw] OR ionamine[tw] OR "1,1 dimethyl 2 phenylethylamine"[tw] OR "2 benzyl 2 propylamine"[tw] OR "alpha,alpha dimethylphenethylamine"[tw] OR "beta aminoisopropylbenzene"[tw] OR "beta phenyltert.butylamine"[tw] OR dimethylphenethylamine[tw] OR dimethylphenetylamine[tw] OR "ex adipos"[tw] OR exadipos[tw] OR fastin[tw] OR lomaira[tw] OR "minobese−forte"[tw] OR "obermine black and yellow"[tw] OR "obestin-30"[tw] OR "oby-cap"[tw] OR "oby-trim"[tw] OR "ona-mast"[tw] OR panbesy[tw] OR "panbesyl nyscaps"[tw] OR phentercot[tw] OR phentermide[tw] OR phentermin[tw] OR phentrol[tw] OR "pro-fast"[tw] OR reducyl[tw] OR redusa[tw] OR suprenza[tw] OR "t-diet"[tw] OR terbutylamine[tw] OR umine[tw] OR wilpo[tw] OR zantryl[tw] OR "Diethylpropion"[Mesh] OR 19v2pl39ng[tw] OR "2-diethylaminopropiophenone"[tw] OR "90-84-6 (diethylpropion)"[tw] OR amfepramon[tw] OR amfepramone[tw] OR anorex[tw] OR delgamer[tw] OR diethylpropion[tw] OR "dietil retard"[tw] OR "dietil-retard"[tw] OR dietilretard[tw] OR "ifa norex"[tw] OR lipomin[tw] OR maruate[tw] OR moderatan[tw] OR neobes[tw] OR nobesine[tw] OR phepranon[tw] OR prefamone[tw] OR propion[tw] OR q94yyu22b8[tw] OR regenon[tw] OR regibon[tw] OR tenuate[tw] OR tepanil[tw] OR "1 benzoyltriethylamine"[tw] OR "2 diethylaminopropiophenone"[tw] OR adiposon[tw] OR alipid[tw] OR "alpha benzoyltriethylamine hydrochloride"[tw] OR "alpha diethylaminopropiophenone"[tw] OR amphepramon[tw] OR anorexil[tw] OR apisate[tw] OR atractil[tw] OR derfon[tw] OR diepropon[tw] OR "diethyl propion"[tw] OR diethylaminopropiophenone[tw] OR diethylpropion[tw] OR diethylpropione[tw] OR dobesin[tw] OR dualid[tw] OR elontril[tw] OR "linea magerion"[tw] OR magrene[tw] OR moludar[tw] OR obesitex[tw] OR prothin[tw] OR sacin[tw] OR "t 712"[tw] OR tylinal[tw] OR voxra[tw] OR zyntabac[tw]) NOT (("Child"[Mesh] OR "Infant"[Mesh] OR "Adolescent"[Mesh]) NOT "Adult"[Mesh]) AND ("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tw] OR randomised[tw] OR placebo[tw] OR "drug therapy"[sh] OR randomly[tw] OR trial[tw] OR groups[tw]) NOT ("Animals"[Mesh] NOT "Humans"[Mesh])

PICO 9

("Obesity"[Mesh] OR "Weight Loss"[Mesh] OR "Overweight"[Mesh] OR obes*[tw] OR "body mass ind*"[tw] OR adiposity[tw] OR overweight[tw] OR "over weight"[tw] OR "overload syndrome*"[tw] OR "over eat*"[tw] OR overfeed*[tw] OR "over feed*"[tw] OR overfed[tw] OR "over fed"[tw] OR "weight cycling"[tw] OR "skinfold thickness"[tw] OR antiobesity[tw] OR "anti-obesity"[tw] OR obesitas[tw] OR bodyweight[tw] OR "body weight"[tw] OR adipositas[tw] OR BMI[tw]) AND (((plenity[tw] OR gelesis100[tw] OR "gelesis 100"[tw] OR "gelesis-100"[tw] OR gs100[tw] OR "gs 100"[tw] OR "gs-100"[tw] OR (superabsorbent[tw] AND hydrogel*[tw]) OR (oral[tw] AND hydrogel*[tw]) OR OSH[tw] OR "carboxymethylcellulose-citric"[tw] OR (carboxymethylcellulose[tw] AND citric[tw]) OR (carboxymethyl[tw] AND cellulose[tw] AND citric[tw]))) OR ((oral[tw] OR superabsorbent[tw]) AND "Hydrogels"[Mesh])) NOT (("Child"[Mesh] OR "Infant"[Mesh] OR "Adolescent"[Mesh]) NOT "Adult"[Mesh]) AND ("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tw] OR randomised[tw] OR placebo[tw] OR "drug therapy"[sh] OR randomly[tw] OR trial[tw] OR groups[tw]) NOT ("Animals"[Mesh] NOT "Humans"[Mesh])

Supplementary Figure 2. PRISMA Flow Diagram

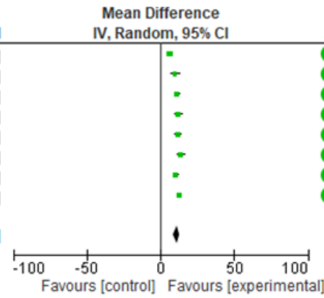


Supplementary Figure 3. Benefits and harms of semaglutide 2.4 mg over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; **(B)** Mean difference in weight loss (in kg); **(C)** Proportion of patients achieving at least 5% total body weight loss; **(D)** Proportion of patients achieving at least 10% total body weight loss; **(E)** Proportion of patients achieving at least 15% total body weight loss; **(F)** Serious adverse events; **(G)** Discontinuation of treatment due to adverse events

A.

	Experimental			Control				Mean Difference	Risk of Bias						
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**	**A**	**B**	**C**	**D**	**E**	**F**	**G**
Davies 2021	9.64	8.0399	404	3.42	8.0299	403	13.7%	6.22 [5.11, 7.33]	+	+	+	+	+	+	+
Flint 2021	10.83	5.8856	33	1.15	5.8856	30	11.0%	9.68 [6.77, 12.59]	+	?	+	?	?	+	+
Kadowaki 2022	13.2	6	199	2.14	8.2957	101	12.8%	11.06 [9.24, 12.88]	+	+	+	+	+	+	+
Newsome 2021	12.51	7.639	81	0.61	7.639	80	12.0%	11.90 [9.54, 14.26]	+	+	+	+	+	+	+
O'Neil 2018	13.8	8.3826	102	2.3	8.6298	136	12.3%	11.50 [9.32, 13.68]	+	+	+	+	+	+	+
Rubino 2022	15.3	9.6419	126	1.9	9.736	85	11.5%	13.40 [10.73, 16.07]	+	+	+	+	+	+	+
Wadden 2021	16	10.111	407	5.7	10.111	204	13.0%	10.30 [8.60, 12.00]	+	+	+	+	+	+	+
Wilding 2021	14.85	9.9103	1306	2.41	9.9103	655	13.8%	12.44 [11.51, 13.37]	+	+	+	+	+	+	+
Total (95% CI)			**2658**			**1694**	**100.0%**	**10.76 [8.73, 12.80]**							

Heterogeneity: Tau² = 7.56; Chi² = 81.22, df = 7 (P < 0.00001); I² = 91%
Test for overall effect: Z = 10.36 (P < 0.00001)

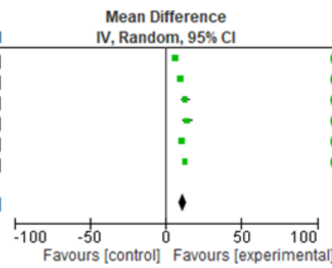


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

B.

	Experimental			Control				Mean Difference	Risk of Bias						
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**	**A**	**B**	**C**	**D**	**E**	**F**	**G**
Davies 2021	9.7	8.0399	404	3.5	8.0299	403	17.7%	6.20 [5.09, 7.31]	+	+	+	+	+	+	+
Kadowaki 2022	11.25	6.4891	199	1.7	6.5324	101	17.2%	9.55 [7.99, 11.11]	+	+	+	+	+	+	+
O'Neil 2018	15.15	9.2915	102	2.48	9.5628	136	15.8%	12.67 [10.25, 15.09]	+	+	+	+	+	+	+
Rubino 2022	15.35	11.0598	126	1.55	10.895	85	14.7%	13.80 [10.78, 16.82]	+	+	+	+	+	+	+
Wadden 2021	16.8	10.7057	407	6.2	10.7057	204	16.8%	10.60 [8.80, 12.40]	+	+	+	+	+	+	+
Wilding 2021	15.3	10.6563	1306	2.6	10.6563	655	17.8%	12.70 [11.70, 13.70]	+	+	+	+	+	+	+
Total (95% CI)			**2544**			**1584**	**100.0%**	**10.81 [8.19, 13.43]**							

Heterogeneity: Tau² = 9.77; Chi² = 84.35, df = 5 (P < 0.00001); I² = 94%
Test for overall effect: Z = 8.09 (P < 0.00001)

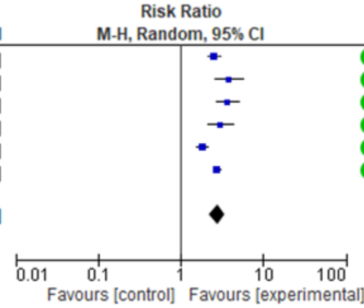


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

C.

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G
Davies 2021	267	403	107	402	18.9%	2.49 [2.09, 2.97]	+	+	+	+	+	+	+
Kadowaki 2022	160	199	21	101	12.8%	3.87 [2.63, 5.69]	+	+	+	+	+	+	+
O'Neil 2018	85	102	31	136	14.7%	3.66 [2.65, 5.04]	+	+	+	+	+	+	+
Rubino 2022	102	126	23	85	13.6%	2.99 [2.09, 4.28]	+	+	+	+	+	+	+
Wadden 2021	352	407	97	204	19.6%	1.82 [1.57, 2.11]	+	+	+	+	+	+	+
Wilding 2021	1128	1306	206	655	20.4%	2.75 [2.45, 3.08]	+	+	+	+	+	+	+
Total (95% CI)		**2543**		**1583**	**100.0%**	**2.74 [2.21, 3.40]**							
Total events	2094		485										

Heterogeneity: Tau² = 0.06; Chi² = 32.15, df = 5 (P < 0.00001); I² = 84%
Test for overall effect: Z = 9.14 (P < 0.00001)

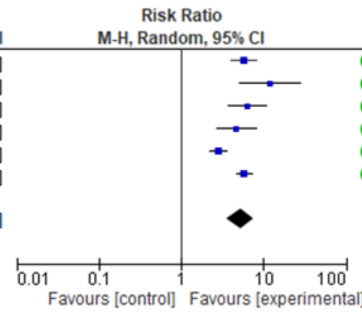


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

D.

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F	G
Davies 2021	177	403	31	402	18.4%	5.70 [3.99, 8.13]	+	+	+	+	+	+	+
Kadowaki 2022	117	199	5	101	10.2%	11.88 [5.01, 28.13]	+	+	+	+	+	+	+
O'Neil 2018	66	102	14	136	15.5%	6.29 [3.75, 10.53]	+	+	+	+	+	+	+
Rubino 2022	83	126	12	85	15.1%	4.67 [2.72, 8.00]	+	+	+	+	+	+	+
Wadden 2021	306	407	55	204	20.3%	2.79 [2.21, 3.52]	+	+	+	+	+	+	+
Wilding 2021	902	1306	78	655	20.6%	5.80 [4.69, 7.17]	+	+	+	+	+	+	+
Total (95% CI)		**2543**		**1583**	**100.0%**	**5.25 [3.61, 7.64]**							
Total events	1651		195										

Heterogeneity: Tau² = 0.17; Chi² = 31.25, df = 5 (P < 0.00001); I² = 84%
Test for overall effect: Z = 8.66 (P < 0.00001)

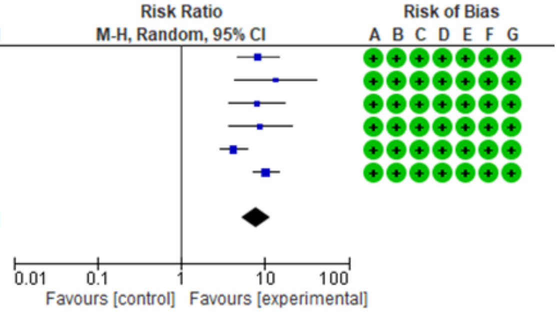


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

E.

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Davies 2021	100	403	12	402	17.9%	8.31 [4.64, 14.89]
Kadowaki 2022	79	199	3	101	9.0%	13.37 [4.33, 41.28]
O'Neil 2018	42	102	7	136	14.3%	8.00 [3.75, 17.07]
Rubino 2022	65	126	5	85	12.4%	8.77 [3.69, 20.87]
Wadden 2021	227	407	27	204	23.0%	4.21 [2.93, 6.05]
Wilding 2021	659	1306	32	655	23.5%	10.33 [7.34, 14.54]
Total (95% CI)		**2543**		**1583**	**100.0%**	**7.82 [5.19, 11.76]**
Total events	1172		86			

Heterogeneity: Tau² = 0.16; Chi² = 14.92, df = 5 (P = 0.01); I² = 66%
Test for overall effect: Z = 9.86 (P < 0.00001)

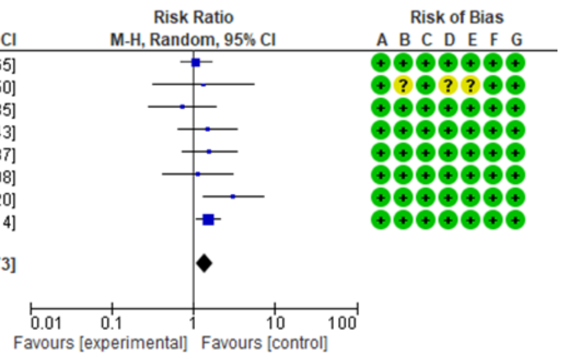


Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

F.

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Davies 2021	40	403	37	402	25.4%	1.08 [0.70, 1.65]
Flint 2021	4	33	3	33	2.5%	1.33 [0.32, 5.50]
Kadowaki 2022	10	199	7	101	5.7%	0.73 [0.28, 1.85]
Newsome 2021	12	81	8	80	7.1%	1.48 [0.64, 3.43]
O'Neil 2018	13	102	11	136	8.5%	1.58 [0.74, 3.37]
Rubino 2022	10	126	6	85	5.3%	1.12 [0.42, 2.98]
Wadden 2021	37	407	6	204	7.0%	3.09 [1.33, 7.20]
Wilding 2021	128	1306	42	655	38.5%	1.53 [1.09, 2.14]
Total (95% CI)		**2657**		**1696**	**100.0%**	**1.38 [1.10, 1.73]**
Total events	254		120			

Heterogeneity: Tau² = 0.01; Chi² = 7.33, df = 7 (P = 0.40); I² = 5%
Test for overall effect: Z = 2.79 (P = 0.005)



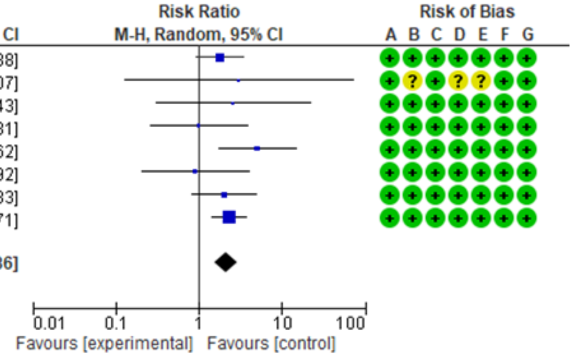
Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

G.

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Davies 2021	25	403	14	402	23.6%	1.78 [0.94, 3.38]
Flint 2021	1	33	0	33	1.0%	3.00 [0.13, 71.07]
Kadowaki 2022	5	199	1	101	2.1%	2.54 [0.30, 21.43]
Newsome 2021	4	81	4	80	5.3%	0.99 [0.26, 3.81]
O'Neil 2018	15	102	4	136	8.4%	5.00 [1.71, 14.62]
Rubino 2022	4	126	3	85	4.4%	0.90 [0.21, 3.92]
Wadden 2021	24	407	6	204	12.5%	2.00 [0.83, 4.83]
Wilding 2021	92	1306	20	655	42.8%	2.31 [1.44, 3.71]
Total (95% CI)		**2657**		**1696**	**100.0%**	**2.10 [1.54, 2.86]**
Total events	170		52			

Heterogeneity: Tau² = 0.00; Chi² = 5.50, df = 7 (P = 0.60); I² = 0%

Test for overall effect: Z = 4.67 (P < 0.00001)



Risk of bias legend

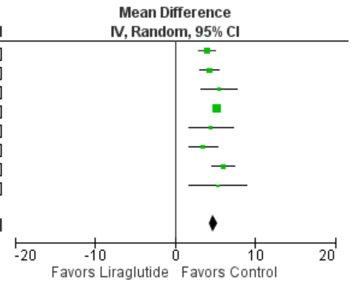
(**A**) Random sequence generation (selection bias)
(**B**) Allocation concealment (selection bias)
(**C**) Blinding of participants and personnel (performance bias)
(**D**) Blinding of outcome assessment (detection bias)
(**E**) Incomplete outcome data (attrition bias)
(**F**) Selective reporting (reporting bias)
(**G**) Other bias

Supplementary Figure 4. Benefits and harms of liraglutide 3.0 mg over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; **(B)** Mean difference in weight loss (in kg); **(C)** Proportion of patients achieving at least 5% total body weight loss; **(D)** Proportion of patients achieving at least 10% total body weight loss; **(E)** Proportion of patients achieving at least 15% total body weight loss; **(F)** Serious adverse events; **(G)** Discontinuation of treatment due to adverse events

A.

	Liraglutide			Control				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
Davies (SCALE Diabetes) 2015	6	6.6296	412	2	6.6296	211	17.2%	4.00 [2.90, 5.10]
Garvey (SCALE Insulin) 2020	5.8	6.0919	198	1.5	6.0919	198	15.4%	4.30 [3.10, 5.50]
O'Neil 2018	7.8	8.6266	103	2.3	8.6298	136	6.0%	5.50 [3.29, 7.71]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	7.8	7.3333	2481	2.6	7.3333	1239	33.9%	5.20 [4.70, 5.70]
Rubino (STEP 8) 2022	6.4	9.8302	117	1.9	9.3141	78	4.1%	4.50 [1.77, 7.23]
Wadden (SCALE IBT) 2020	7.5	7.711	142	4	7.711	140	8.4%	3.50 [1.70, 5.30]
Wadden (SCALE Maintenance) 2013	6.2	7.3	207	0.2	7	206	12.7%	6.00 [4.62, 7.38]
Wadden 2019	11.5	9.1924	50	6.1	9.1924	50	2.4%	5.40 [1.80, 9.00]
Total (95% CI)			**3710**			**2258**	**100.0%**	**4.81 [4.23, 5.39]**

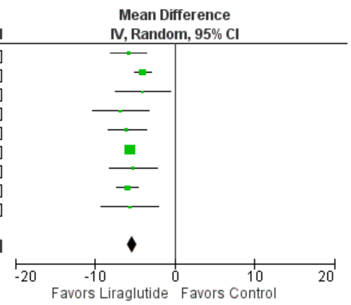
Heterogeneity: Tau² = 0.19; Chi² = 10.07, df = 7 (P = 0.18); I² = 31%
Test for overall effect: Z = 16.30 (P < 0.00001)



B.

	Liraglutide			Control				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
Astrup 2012	-7.8	7.7322	92	-2	7.7322	98	6.5%	-5.80 [-8.00, -3.60]
Davies (SCALE Diabetes) 2015	-7.1	6.2	412	-3.1	5.9	211	21.6%	-4.00 [-5.00, -3.00]
Gudbergsen 2021	-2.8	11.234	80	1.2	10.5028	76	2.9%	-4.00 [-7.41, -0.59]
Lundgren 2021	-0.7	8.7037	49	6.1	9.0519	49	2.8%	-6.80 [-10.32, -3.28]
O'Neil 2018	-8.47	9.4385	103	-2.48	9.5628	136	5.5%	-5.99 [-8.42, -3.56]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	-8.4	7.3	2437	-2.8	6.5	1225	40.2%	-5.60 [-6.07, -5.13]
Rubino (STEP 8) 2022	-6.8	10.9225	117	-1.6	10.2011	78	3.7%	-5.20 [-8.21, -2.19]
Wadden (SCALE Maintenance) 2013	-6	7.3	207	-0.1	6.9	206	14.1%	-5.90 [-7.27, -4.53]
Wadden 2019	-12.2	9.1924	50	-6.6	9.1924	50	2.6%	-5.60 [-9.20, -2.00]
Total (95% CI)			**3547**			**2129**	**100.0%**	**-5.30 [-5.90, -4.70]**

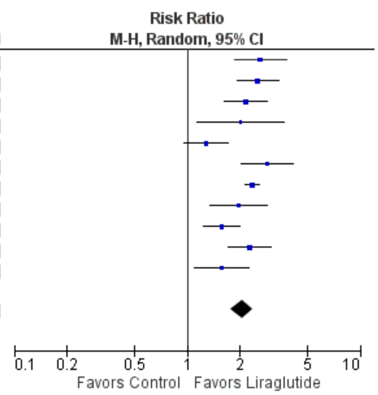
Heterogeneity: Tau² = 0.18; Chi² = 10.45, df = 8 (P = 0.24); I² = 23%
Test for overall effect: Z = 17.29 (P < 0.00001)



C.

	Liraglutide		Control			Risk Ratio
Study or Subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Random, 95% CI**
Astrup 2012	68	93	27	98	8.3%	2.65 [1.88, 3.74]
Davies (SCALE Diabetes) 2015	230	423	45	212	9.9%	2.56 [1.95, 3.37]
Garvey (SCALE Insulin) 2020	103	198	47	198	9.7%	2.19 [1.65, 2.91]
Gudbergsen 2021	28	80	13	76	4.5%	2.05 [1.15, 3.65]
Lundgren 2021	36	49	28	49	9.4%	1.29 [0.96, 1.73]
O'Neil 2018	68	103	31	136	8.4%	2.90 [2.06, 4.06]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	1549	2487	326	1244	14.2%	2.38 [2.15, 2.62]
Rubino (STEP 8) 2022	68	127	23	85	7.4%	1.98 [1.35, 2.91]
Wadden (SCALE IBT) 2020	87	142	54	140	10.6%	1.59 [1.24, 2.03]
Wadden (SCALE Maintenance) 2013	107	212	46	210	9.6%	2.30 [1.73, 3.07]
Wadden 2019	35	50	22	50	7.9%	1.59 [1.11, 2.28]
Total (95% CI)		**3964**		**2498**	**100.0%**	**2.09 [1.80, 2.42]**
Total events	2379		662			

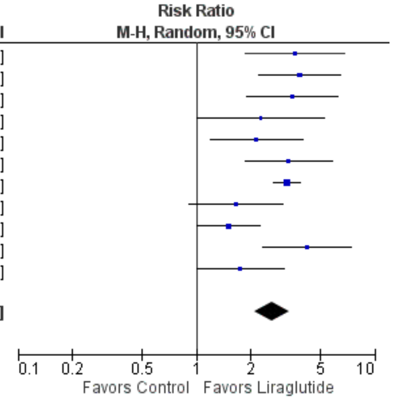
Heterogeneity: Tau² = 0.04; Chi² = 31.01, df = 10 (P = 0.0006); I² = 68%
Test for overall effect: Z = 9.77 (P < 0.00001)



D.

Study or Subgroup	Liraglutide Events	Liraglutide Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	34	93	10	98	7.3%	3.58 [1.88, 6.83]
Davies (SCALE Diabetes) 2015	104	412	14	211	9.1%	3.80 [2.23, 6.48]
Garvey (SCALE Insulin) 2020	45	198	13	198	8.2%	3.46 [1.93, 6.21]
Gudbergsen 2021	17	80	7	76	5.3%	2.31 [1.01, 5.25]
Lundgren 2021	24	49	11	49	8.1%	2.18 [1.20, 3.95]
O'Neil 2018	35	103	14	136	8.6%	3.30 [1.88, 5.80]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	798	2487	124	1244	16.9%	3.22 [2.70, 3.84]
Rubino (STEP 8) 2022	30	127	12	85	7.8%	1.67 [0.91, 3.08]
Wadden (SCALE IBT) 2020	43	142	28	140	11.4%	1.51 [1.00, 2.29]
Wadden (SCALE Maintenance) 2013	55	212	13	210	8.4%	4.19 [2.36, 7.44]
Wadden 2019	23	50	13	50	8.7%	1.77 [1.01, 3.08]
Total (95% CI)		**3953**		**2497**	**100.0%**	**2.67 [2.14, 3.34]**
Total events	1208		259			

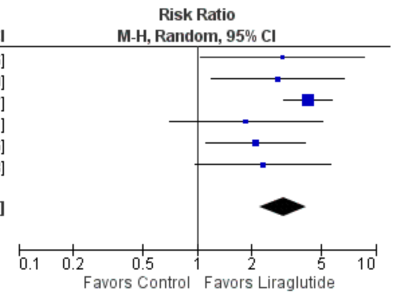
Heterogeneity: Tau² = 0.07; Chi² = 22.01, df = 10 (P = 0.02); I² = 55%
Test for overall effect: Z = 8.61 (P < 0.00001)



E.

Study or Subgroup	Liraglutide Events	Liraglutide Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Lundgren 2021	12	49	4	49	7.5%	3.00 [1.04, 8.66]
O'Neil 2018	15	103	7	136	10.8%	2.83 [1.20, 6.69]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	358	2487	43	1244	45.0%	4.16 [3.06, 5.67]
Rubino (STEP 8) 2022	14	127	5	85	8.5%	1.87 [0.70, 5.01]
Wadden (SCALE IBT) 2020	26	142	12	140	17.6%	2.14 [1.12, 4.06]
Wadden 2019	14	50	6	50	10.6%	2.33 [0.98, 5.58]
Total (95% CI)		**2958**		**1704**	**100.0%**	**3.04 [2.25, 4.12]**
Total events	439		77			

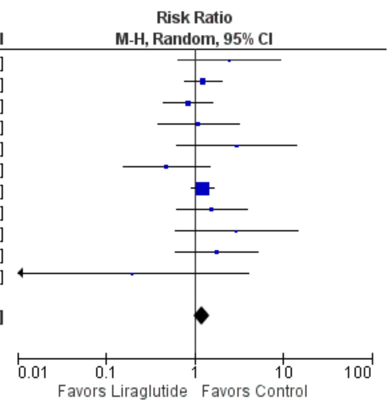
Heterogeneity: Tau² = 0.03; Chi² = 6.15, df = 5 (P = 0.29); I² = 19%
Test for overall effect: Z = 7.20 (P < 0.00001)



F.

Study or Subgroup	Liraglutide Events	Liraglutide Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	7	93	3	98	2.4%	2.46 [0.66, 9.23]
Davies (SCALE Diabetes) 2015	52	423	21	212	18.0%	1.24 [0.77, 2.00]
Garvey (SCALE Insulin) 2020	16	198	19	198	10.2%	0.84 [0.45, 1.59]
Gudbergsen 2021	7	80	6	76	3.8%	1.11 [0.39, 3.15]
Lundgren 2021	6	49	2	49	1.7%	3.00 [0.64, 14.14]
O'Neil 2018	4	103	11	136	3.3%	0.48 [0.16, 1.46]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	154	2487	62	1244	50.1%	1.24 [0.93, 1.66]
Rubino (STEP 8) 2022	14	127	6	85	4.9%	1.56 [0.62, 3.90]
Wadden (SCALE IBT) 2020	6	142	2	140	1.6%	2.96 [0.61, 14.41]
Wadden (SCALE Maintenance) 2013	9	212	5	210	3.6%	1.78 [0.61, 5.23]
Wadden 2019	0	50	2	50	0.5%	0.20 [0.01, 4.06]
Total (95% CI)		**3964**		**2498**	**100.0%**	**1.22 [1.00, 1.50]**
Total events	275		139			

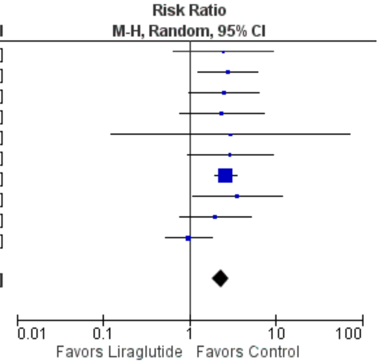
Heterogeneity: Tau² = 0.00; Chi² = 9.76, df = 10 (P = 0.46); I² = 0%
Test for overall effect: Z = 1.96 (P = 0.05)



G.

Study or Subgroup	Liraglutide Events	Liraglutide Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	7	93	3	98	2.8%	2.46 [0.66, 9.23]
Davies (SCALE Diabetes) 2015	39	423	7	212	8.0%	2.79 [1.27, 6.14]
Garvey (SCALE Insulin) 2020	15	198	6	198	5.8%	2.50 [0.99, 6.31]
Gudbergsen 2021	10	80	4	76	4.0%	2.38 [0.78, 7.25]
Lundgren 2021	1	49	0	49	0.5%	3.00 [0.13, 71.89]
O'Neil 2018	9	103	4	136	3.8%	2.97 [0.94, 9.38]
Pi-Sunyer (SCALE Obesity and Prediabetes) 2015	246	2487	47	1244	53.5%	2.62 [1.93, 3.55]
Rubino (STEP 8) 2022	16	127	3	85	3.4%	3.57 [1.07, 11.88]
Wadden (SCALE IBT) 2020	12	142	6	140	5.5%	1.97 [0.76, 5.11]
Wadden (SCALE Maintenance) 2013	18	212	18	210	12.7%	0.99 [0.53, 1.85]
Total (95% CI)		**3914**		**2448**	**100.0%**	**2.31 [1.85, 2.88]**
Total events	373		98			

Heterogeneity: Tau² = 0.00; Chi² = 8.84, df = 9 (P = 0.45); I² = 0%
Test for overall effect: Z = 7.36 (P < 0.00001)

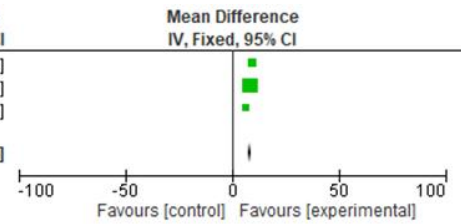


Supplementary Figure 5. Benefits and harms of phentermine-topiramate ER over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; (**B)** Proportion of patients achieving at least 5% total body weight loss; **(C)** Proportion of patients achieving at least 10% total body weight loss; **(D)** Serious adverse events; **(E)** Discontinuation of treatment due to adverse events

A.

	Experimental			Control				Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Fixed, 95% CI
Allison 2011	10.95	8.6381	511	1.55	8.655	513	28.1%	9.40 [8.34, 10.46]
Gadde 2011	9.85	8.8365	994	1.25	8.832	993	52.2%	8.60 [7.82, 9.38]
Garvey 2014	9.4	3.4814	75	2.7	3.7361	55	19.7%	6.70 [5.44, 7.96]
Total (95% CI)			**1580**			**1561**	**100.0%**	**8.45 [7.89, 9.01]**

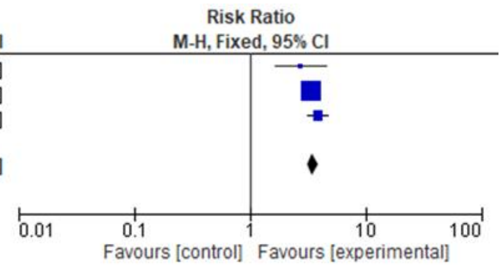
Heterogeneity: Chi² = 10.61, df = 2 (P = 0.005); I² = 81%
Test for overall effect: Z = 29.51 (P < 0.00001)



B.

	Experimental		Control			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Fixed, 95% CI
Garvey 2014	49	75	13	55	4.9%	2.76 [1.67, 4.57]
Gadde 2011	687	994	204	993	66.9%	3.36 [2.96, 3.83]
Allison 2011	332	511	86	513	28.1%	3.88 [3.16, 4.75]
Total (95% CI)		**1580**		**1561**	**100.0%**	**3.48 [3.13, 3.87]**
Total events	1068		303			

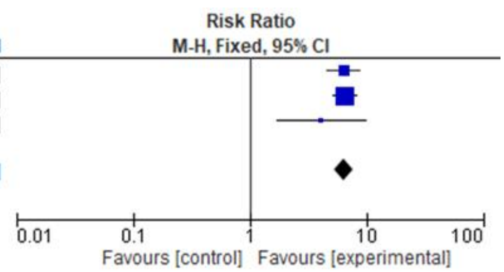
Heterogeneity: Chi² = 2.15, df = 2 (P = 0.34); I² = 7%
Test for overall effect: Z = 22.93 (P < 0.00001)



C.

	Experimental		Control			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Fixed, 95% CI
Allison 2011	235	511	37	513	32.2%	6.38 [4.61, 8.82]
Gadde 2011	467	994	72	993	62.8%	6.48 [5.14, 8.17]
Garvey 2014	28	75	5	55	5.0%	4.11 [1.69, 9.96]
Total (95% CI)		**1580**		**1561**	**100.0%**	**6.33 [5.26, 7.61]**
Total events	730		114			

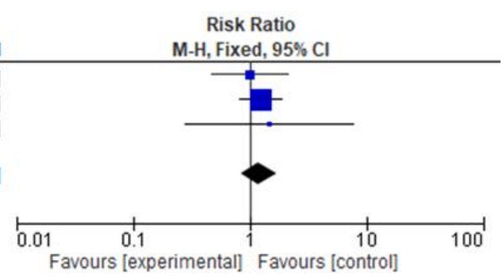
Heterogeneity: Chi² = 0.96, df = 2 (P = 0.62); I² = 0%
Test for overall effect: Z = 19.61 (P < 0.00001)



D.

	Experimental		Control			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Fixed, 95% CI
Allison 2011	13	511	13	513	23.5%	1.00 [0.47, 2.14]
Gadde 2011	50	994	40	993	72.4%	1.25 [0.83, 1.87]
Garvey 2014	4	75	2	55	4.2%	1.47 [0.28, 7.72]
Total (95% CI)		**1580**		**1561**	**100.0%**	**1.20 [0.85, 1.70]**
Total events	67		55			

Heterogeneity: Chi² = 0.31, df = 2 (P = 0.86); I² = 0%
Test for overall effect: Z = 1.02 (P = 0.31)

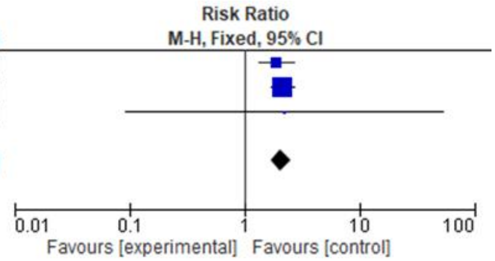


E.

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Allison 2011	82	511	43	513	32.4%	1.91 [1.35, 2.71]
Gadde 2011	192	994	89	993	67.2%	2.16 [1.70, 2.73]
Garvey 2014	1	75	0	55	0.4%	2.21 [0.09, 53.26]
Total (95% CI)		**1580**		**1561**	**100.0%**	**2.08 [1.71, 2.52]**
Total events	275		132			

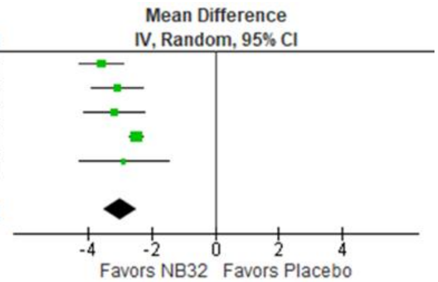
Heterogeneity: Chi² = 0.31, df = 2 (P = 0.86); I² = 0%

Test for overall effect: Z = 7.36 (P < 0.00001)



Supplementary Figure 6. Benefits and harms of naltrexone-bupropion ER over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; **(B)** Mean difference in weight loss (in kg); **(C)** Proportion of patients achieving at least 5% total body weight loss; **(D)** Proportion of patients achieving at least 10% total body weight loss; **(E)** Proportion of patients achieving at least 15% total body weight loss; **(F)** Serious adverse events; **(G)** Discontinuation of treatment due to adverse events

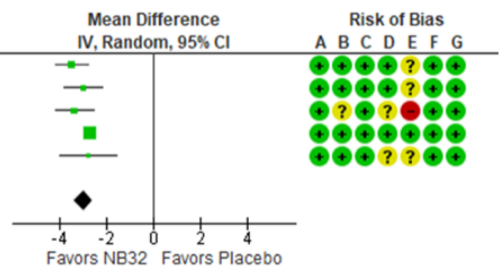
A.



Study or Subgroup	NB 32 Mean	NB 32 SD	NB 32 Total	Placebo Mean	Placebo SD	Placebo Total	Weight	Mean Difference IV, Random, 95% CI
Apovian, 2013	-4.4	5.93	878	-0.8	6.67	495	21.7%	-3.60 [-4.31, -2.89]
Greenway, 2010	-4	7.24	583	-0.9	7.23	581	19.0%	-3.10 [-3.93, -2.27]
Hollander, 2013	-5	4.88	265	-1.8	5.04	159	16.1%	-3.20 [-4.18, -2.22]
Nissen, 2016	-3.6	6.67	4455	-1.1	3.4	4450	33.0%	-2.50 [-2.72, -2.28]
Wadden, 2010	-7.8	9.72	591	-4.9	8.53	202	10.2%	-2.90 [-4.31, -1.49]
Total (95% CI)			**6772**			**5887**	**100.0%**	**-3.01 [-3.54, -2.47]**

Heterogeneity: Tau² = 0.21; Chi² = 11.19, df = 4 (P = 0.02); I² = 64%
Test for overall effect: Z = 10.99 (P < 0.00001)

B.

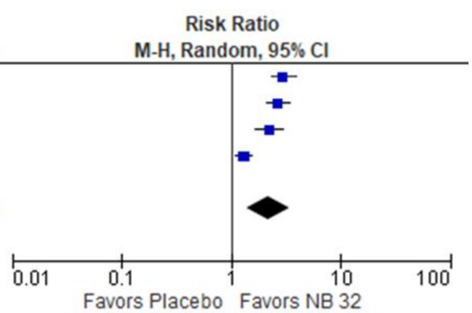


Study or Subgroup	NB 32 Mean	NB 32 SD	NB 32 Total	Placebo Mean	Placebo SD	Placebo Total	Weight	Mean Difference IV, Random, 95% CI
Apovian, 2013	-4.3	5.93	878	-0.8	6.67	495	18.6%	-3.50 [-4.21, -2.79]
Greenway, 2010	-3.9	7.24	583	-0.9	7.23	581	14.9%	-3.00 [-3.83, -2.17]
Hollander, 2013	-5.3	4.88	265	-1.9	3.78	159	14.9%	-3.40 [-4.23, -2.57]
Nissen, 2016	-3.9	6.67	4455	-1.2	3.4	4450	43.5%	-2.70 [-2.92, -2.48]
Wadden, 2010	-7.8	4.86	591	-5	8.53	202	8.0%	-2.80 [-4.04, -1.56]
Total (95% CI)			**6772**			**5887**	**100.0%**	**-3.01 [-3.39, -2.62]**

Heterogeneity: Tau² = 0.07; Chi² = 6.76, df = 4 (P = 0.15); I² = 41%
Test for overall effect: Z = 15.42 (P < 0.00001)

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

C.



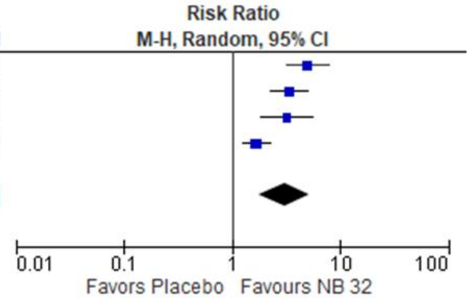
Study or Subgroup	NB 32 Events	NB 32 Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Apovian, 2013	307	878	58	495	24.9%	2.98 [2.31, 3.86]
Greenway, 2010	180	583	67	581	24.9%	2.68 [2.07, 3.46]
Hollander, 2013	141	265	38	159	24.1%	2.23 [1.65, 3.00]
Wadden, 2010	321	591	84	202	26.1%	1.31 [1.09, 1.56]
Total (95% CI)		**2317**		**1437**	**100.0%**	**2.18 [1.41, 3.37]**
Total events	949		247			

Heterogeneity: Tau² = 0.18; Chi² = 38.37, df = 3 (P < 0.00001); I² = 92%
Test for overall effect: Z = 3.52 (P = 0.0004)

D.

	NB 32		Placebo			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Apovian, 2013	187	878	21	495	24.7%	5.02 [3.24, 7.78]
Greenway, 2010	101	583	30	581	25.6%	3.36 [2.27, 4.96]
Hollander, 2013	70	265	13	159	22.3%	3.23 [1.85, 5.65]
Wadden, 2010	201	591	41	202	27.3%	1.68 [1.25, 2.25]
Total (95% CI)		**2317**		**1437**	**100.0%**	**3.04 [1.80, 5.14]**
Total events	559		105			

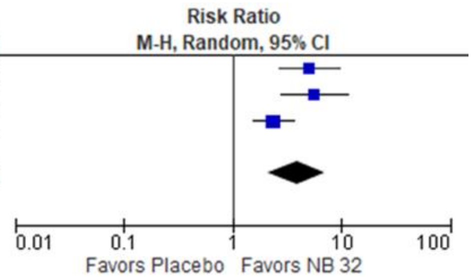
Heterogeneity: Tau² = 0.24; Chi² = 20.02, df = 3 (P = 0.0002); I² = 85%
Test for overall effect: Z = 4.15 (P < 0.0001)



E.

	NB 32		Placebo			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Apovian, 2013	90	878	10	495	31.5%	5.07 [2.66, 9.66]
Greenway, 2010	51	583	9	581	29.5%	5.65 [2.81, 11.36]
Wadden, 2010	138	591	20	202	39.0%	2.36 [1.52, 3.67]
Total (95% CI)		**2052**		**1278**	**100.0%**	**3.88 [2.13, 7.08]**
Total events	279		39			

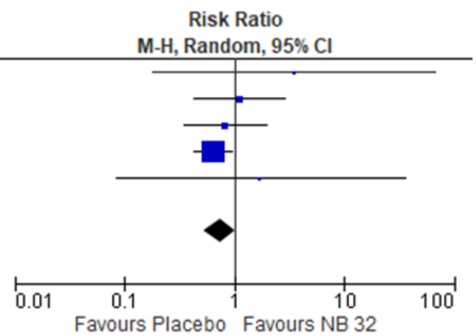
Heterogeneity: Tau² = 0.19; Chi² = 6.22, df = 2 (P = 0.04); I² = 68%
Test for overall effect: Z = 4.44 (P < 0.00001)



F.

	NB 32		Placebo			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Apovian, 2013	3	992	0	492	1.3%	3.48 [0.18, 67.15]
Greenway, 2010	9	583	8	581	12.3%	1.12 [0.44, 2.89]
Hollander, 2013	13	333	8	169	14.8%	0.82 [0.35, 1.95]
Nissen, 2016	40	4455	62	4450	70.4%	0.64 [0.43, 0.96]
Wadden, 2010	2	584	0	200	1.2%	1.72 [0.08, 35.63]
Total (95% CI)		**6947**		**5892**	**100.0%**	**0.74 [0.53, 1.03]**
Total events	67		78			

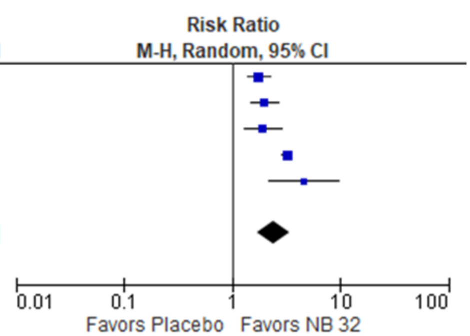
Heterogeneity: Tau² = 0.00; Chi² = 2.62, df = 4 (P = 0.62); I² = 0%
Test for overall effect: Z = 1.78 (P = 0.07)



G.

	NB 32		Placebo			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Apovian, 2013	241	992	68	492	22.7%	1.76 [1.37, 2.25]
Greenway, 2010	112	583	56	581	21.5%	1.99 [1.48, 2.69]
Hollander, 2013	98	333	26	169	19.3%	1.91 [1.29, 2.83]
Nissen, 2016	1253	4455	388	4450	25.0%	3.23 [2.90, 3.59]
Wadden, 2010	94	584	7	200	11.6%	4.60 [2.17, 9.74]
Total (95% CI)		**6947**		**5892**	**100.0%**	**2.39 [1.69, 3.37]**
Total events	1798		545			

Heterogeneity: Tau² = 0.12; Chi² = 30.89, df = 4 (P < 0.00001); I² = 87%
Test for overall effect: Z = 4.95 (P < 0.00001)

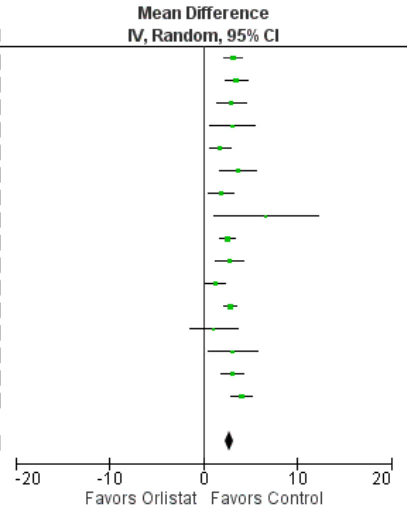


Supplementary Figure 7. Benefits and harms of orlistat over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; **(B)** Mean difference in weight loss (in kg); **(C)** Proportion of patients achieving at least 5% total body weight loss; **(D)** Proportion of patients achieving at least 10% total body weight loss; **(E)** Serious adverse events; **(F)** Discontinuation of treatment due to adverse events

A.

	Orlistat			Control				Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Berne 2005	5	3.2661	111	1.8	4.0717	109	9.2%	3.20 [2.22, 4.18]
Broom 2002	5.8	7.8	259	2.3	6.2	263	7.3%	3.50 [2.29, 4.71]
Davidson 1999	8.8	10.2528	657	5.8	10.4532	223	5.1%	3.00 [1.42, 4.58]
Finer 2000	8.5	9.4393	110	5.4	8.3138	108	2.7%	3.10 [0.74, 5.46]
Hanefeld 2002	5.4	5	189	3.6	5.7	180	8.2%	1.80 [0.70, 2.90]
Hauptman 2000	7.9	11.4482	210	4.2	8.7361	212	3.7%	3.70 [1.76, 5.64]
Hollander 1998	6.2	6.364	162	4.3	6.3048	159	6.1%	1.90 [0.51, 3.29]
James 1997	8.6	11.0304	23	1.9	8.1529	23	0.6%	6.70 [1.09, 12.31]
Kelley 2002	3.76	4.2405	266	1.22	4.9204	269	11.2%	2.54 [1.76, 3.32]
Krempf 2003	5.4	11.1606	346	2.6	9.3541	350	5.4%	2.80 [1.27, 4.33]
Lindgärde 2000	5.9	5.5	190	4.6	5.4	186	8.1%	1.30 [0.20, 2.40]
Miles 2002	4.6	4.7434	250	1.7	3.1875	254	12.0%	2.90 [2.19, 3.61]
Reaven (Cohort 1) 2001	8.1	7.3756	85	7	6.5574	43	2.5%	1.10 [-1.41, 3.61]
Reaven (Cohort 2) 2001	10	8.4261	71	6.9	6.2354	48	2.2%	3.10 [0.46, 5.74]
Rössner 2000	9.7	6.3	242	6.6	6.8	237	7.6%	3.10 [1.93, 4.27]
Sjöström 1998	10.2	7.4081	343	6.1	7.3756	340	8.1%	4.10 [2.99, 5.21]
Total (95% CI)			**3514**			**3004**	**100.0%**	**2.78 [2.36, 3.20]**

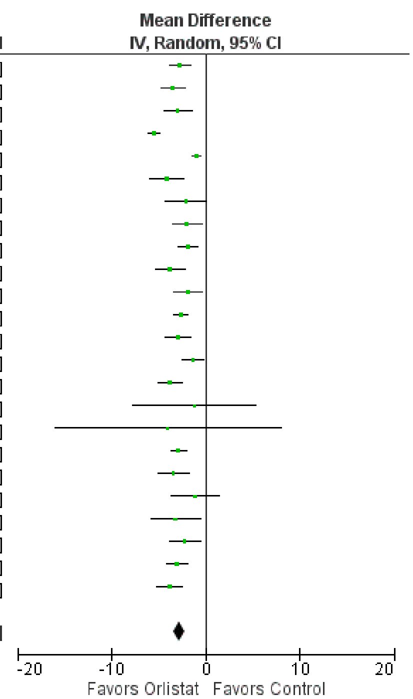
Heterogeneity: Tau² = 0.26; Chi² = 24.49, df = 15 (P = 0.06); I² = 39%
Test for overall effect: Z = 12.88 (P < 0.00001)



B.

	Orlistat			Control				Mean Difference
Study or Subgroup	Mean	SD	Total	Mean	SD	Total	Weight	IV, Random, 95% CI
Bakris 2002	-5.4	6.4	267	-2.7	6.4	265	4.9%	-2.70 [-3.79, -1.61]
Broom 2002	-5.8	8.5	259	-2.3	6.4	263	4.7%	-3.50 [-4.79, -2.21]
Davidson 1999	-8.76	9.4838	657	-5.81	10.0052	223	4.4%	-2.95 [-4.45, -1.45]
Derosa 2002	-12.5	1.3	26	-7	1.2	29	5.4%	-5.50 [-6.16, -4.84]
Derosa 2003	-8.6	1	25	-7.6	0.7	23	5.6%	-1.00 [-1.49, -0.51]
Derosa 2010	85	5.9	113	89.1	7.8	121	4.0%	-4.10 [-5.86, -2.34]
Dixon 2008	-3.1	3.9129	16	-1	2.0203	15	3.5%	-2.10 [-4.27, 0.07]
Finer 2000	-3.29	6.1016	110	-1.31	6.1016	108	4.2%	-1.98 [-3.60, -0.36]
Hanefeld 2002	-5.3	5.1	189	-3.4	5.3	180	5.0%	-1.90 [-2.96, -0.84]
Hauptman 2000	-7.94	8.2601	210	-4.14	8.1537	212	4.3%	-3.80 [-5.37, -2.23]
Hollander 1998	-6.19	6.4912	162	-4.31	7.1874	159	4.4%	-1.88 [-3.38, -0.38]
Kelley 2002	-3.89	4.4036	266	-1.27	4.5923	269	5.3%	-2.62 [-3.38, -1.86]
Krempf 2003	-5.3	9.3005	346	-2.4	9.3541	350	4.5%	-2.90 [-4.29, -1.51]
Lindgärde 2000	-5.6	5.2	190	-4.3	5.9	186	4.9%	-1.30 [-2.43, -0.17]
Lucas 2003	-9.9	6.4	256	-6.1	6.8557	188	4.7%	-3.80 [-5.05, -2.55]
Mathus-Vliegen 2004	-9.1	4.6433	11	-7.9	9	9	0.8%	-1.20 [-7.69, 5.29]
Mathus-Vliegen 2006	90.7	14	10	94.7	12.7	9	0.3%	-4.00 [-16.01, 8.01]
Miles 2002	-4.7	4.7434	250	-1.8	4.7812	254	5.3%	-2.90 [-3.73, -2.07]
Poston 2006	-1.7	6.3	82	1.7	4.2	85	4.2%	-3.40 [-5.03, -1.77]
Reaven (Cohort 1) 2001	-8.5	7.3756	85	-7.4	6.5574	43	3.1%	-1.10 [-3.61, 1.41]
Reaven (Cohort 2) 2001	-9.5	8.4261	71	-6.3	6.2354	48	2.9%	-3.20 [-5.84, -0.56]
Richelsen 2007	-9.4	8.3	153	-7.2	6.3	156	4.2%	-2.20 [-3.85, -0.55]
Rössner 2000	-9.4	6.4	242	-6.4	6.7	237	4.8%	-3.00 [-4.17, -1.83]
Swinburn 2005	-4.7	7.7	170	-0.9	4.2	169	4.6%	-3.80 [-5.12, -2.48]
Total (95% CI)			**4166**			**3601**	**100.0%**	**-2.81 [-3.45, -2.17]**

Heterogeneity: Tau² = 1.83; Chi² = 140.95, df = 23 (P < 0.00001); I² = 84%
Test for overall effect: Z = 8.64 (P < 0.00001)

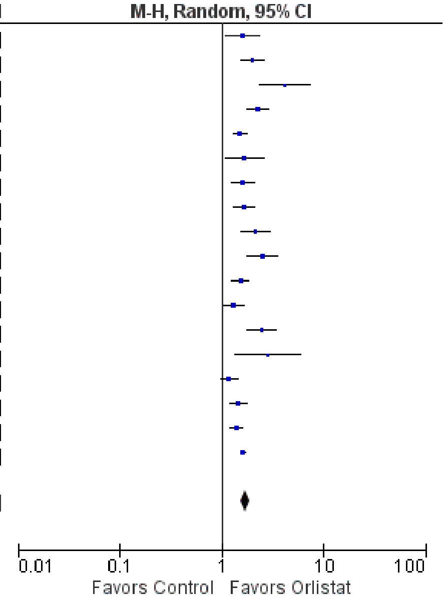


C.

Study or Subgroup	Orlistat Events	Orlistat Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	42	95	27	98	3.7%	1.60 [1.08, 2.38]
Bakris 2002	122	278	60	276	5.6%	2.02 [1.56, 2.62]
Berne 2005	51	111	12	109	2.2%	4.17 [2.36, 7.38]
Broom 2002	147	265	65	266	6.0%	2.27 [1.79, 2.88]
Davidson 1999	439	668	98	224	7.5%	1.50 [1.28, 1.76]
Finer 2000	40	114	24	114	3.3%	1.67 [1.08, 2.57]
Hanefeld 2002	100	195	59	188	5.8%	1.63 [1.27, 2.10]
Hauptman 2000	106	210	65	212	5.9%	1.65 [1.29, 2.10]
Hollander 1998	80	163	36	159	4.6%	2.17 [1.56, 3.01]
Kelley 2002	90	274	36	276	4.3%	2.52 [1.78, 3.57]
Krempf 2003	170	346	112	350	7.0%	1.54 [1.27, 1.85]
Lindgärde 2000	103	190	76	186	6.4%	1.33 [1.07, 1.65]
Miles 2002	99	255	41	261	4.7%	2.47 [1.79, 3.41]
Poston 2006	22	82	8	85	1.4%	2.85 [1.35, 6.04]
Richelsen 2007	102	153	87	156	7.1%	1.20 [1.00, 1.43]
Rössner 2000	153	244	104	243	7.2%	1.47 [1.23, 1.74]
Sjöström 1998	236	345	168	343	8.1%	1.40 [1.23, 1.59]
Torgerson (XENDOS) 2004	1201	1650	746	1655	9.1%	1.61 [1.52, 1.72]
Total (95% CI)		**5638**		**5201**	**100.0%**	**1.71 [1.55, 1.88]**
Total events	3303		1824			

Heterogeneity: Tau² = 0.03; Chi² = 62.65, df = 17 (P < 0.00001); I² = 73%
Test for overall effect: Z = 10.76 (P < 0.00001)

Risk Ratio M-H, Random, 95% CI

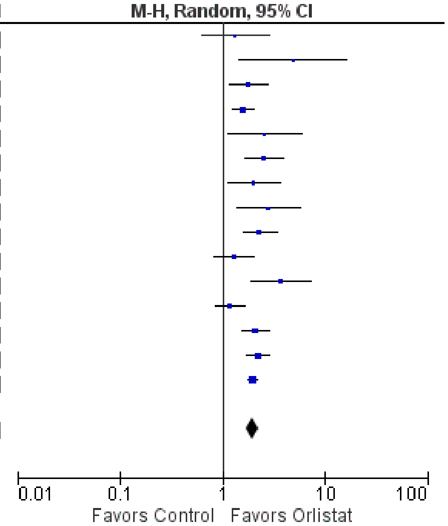


D.

Study or Subgroup	Orlistat Events	Orlistat Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	13	95	10	98	2.6%	1.34 [0.62, 2.91]
Berne 2005	15	111	3	109	1.2%	4.91 [1.46, 16.48]
Broom 2002	52	265	29	266	6.7%	1.80 [1.18, 2.74]
Davidson 1999	260	668	55	224	11.4%	1.59 [1.24, 2.03]
Finer 2000	18	114	7	114	2.3%	2.57 [1.12, 5.92]
Hauptman 2000	60	210	24	212	6.4%	2.52 [1.64, 3.89]
Hollander 1998	29	163	14	159	4.0%	2.02 [1.11, 3.68]
Kelley 2002	28	274	10	276	3.1%	2.82 [1.40, 5.69]
Krempf 2003	75	346	33	350	7.5%	2.30 [1.57, 3.37]
Lindgärde 2000	36	190	27	186	6.0%	1.31 [0.83, 2.06]
Miles 2002	36	255	10	261	3.3%	3.68 [1.87, 7.27]
Richelsen 2007	52	153	45	156	8.8%	1.18 [0.85, 1.64]
Rössner 2000	93	244	44	243	9.4%	2.10 [1.54, 2.87]
Sjöström 1998	134	345	60	343	10.8%	2.22 [1.70, 2.89]
Torgerson (XENDOS) 2004	676	1650	344	1655	16.4%	1.97 [1.76, 2.20]
Total (95% CI)		**5083**		**4652**	**100.0%**	**1.94 [1.70, 2.22]**
Total events	1577		715			

Heterogeneity: Tau² = 0.03; Chi² = 25.99, df = 14 (P = 0.03); I² = 46%
Test for overall effect: Z = 9.57 (P < 0.00001)

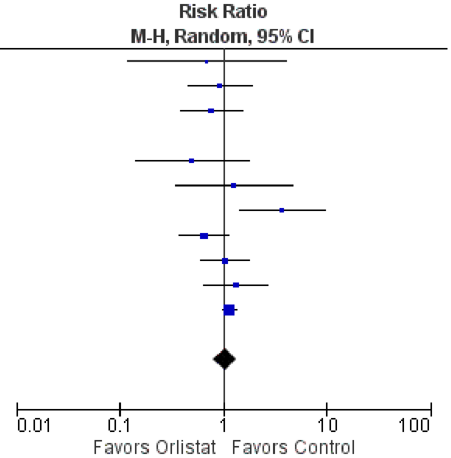
Risk Ratio M-H, Random, 95% CI



E.

Study or Subgroup	Orlistat Events	Orlistat Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	2	95	3	98	1.9%	0.69 [0.12, 4.02]
Bakris 2002	14	278	15	276	9.2%	0.93 [0.46, 1.88]
Broom 2002	13	265	17	266	9.3%	0.77 [0.38, 1.55]
Derosa 2003	0	27	0	23		Not estimable
James 1997	3	23	6	23	3.5%	0.50 [0.14, 1.76]
Krempf 2003	5	346	4	350	3.3%	1.26 [0.34, 4.67]
Lindgärde 2000	19	190	5	186	5.6%	3.72 [1.42, 9.76]
Richelsen 2007	18	153	28	156	13.2%	0.66 [0.38, 1.13]
Sjöström 1998	25	345	24	343	13.4%	1.04 [0.60, 1.78]
Swinburn 2005	16	170	12	169	9.0%	1.33 [0.65, 2.72]
Torgerson (XENDOS) 2004	247	1650	215	1655	31.5%	1.15 [0.97, 1.37]
Total (95% CI)		**3542**		**3545**	**100.0%**	**1.04 [0.81, 1.33]**
Total events	362		329			

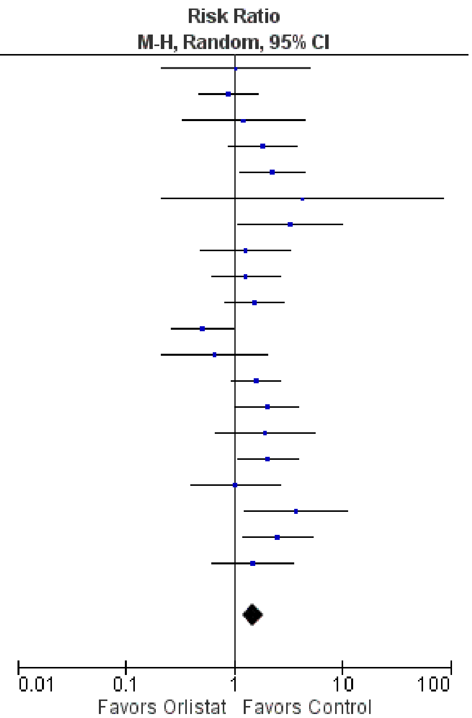
Heterogeneity: Tau² = 0.04; Chi² = 13.21, df = 9 (P = 0.15); I² = 32%
Test for overall effect: Z = 0.30 (P = 0.77)



F.

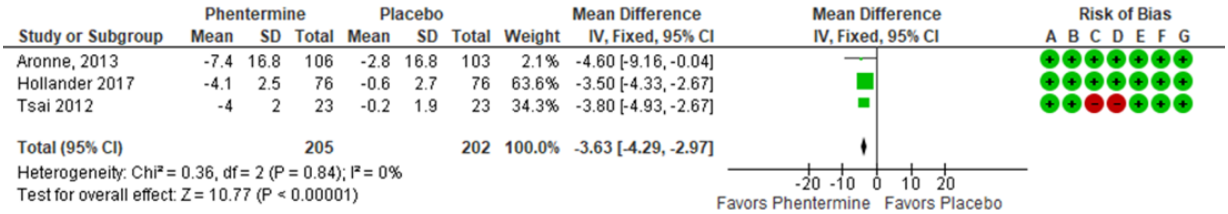
Study or Subgroup	Orlistat Events	Orlistat Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Astrup 2012	3	95	3	98	1.8%	1.03 [0.21, 4.98]
Bakris 2002	18	278	20	276	7.5%	0.89 [0.48, 1.65]
Berne 2005	5	111	4	109	2.5%	1.23 [0.34, 4.45]
Broom 2002	20	265	11	266	6.2%	1.83 [0.89, 3.73]
Davidson 1999	61	668	9	224	6.6%	2.27 [1.15, 4.50]
Derosa 2003	2	27	0	23	0.5%	4.29 [0.22, 84.97]
Derosa 2010	13	126	4	128	3.3%	3.30 [1.11, 9.85]
Finer 2000	9	114	7	114	4.1%	1.29 [0.50, 3.33]
Hanefeld 2002	16	195	12	188	6.1%	1.29 [0.62, 2.64]
Hauptman 2000	23	210	15	212	7.4%	1.55 [0.83, 2.88]
Hollander 1998	12	163	23	159	6.8%	0.51 [0.26, 0.99]
James 1997	4	23	6	23	3.1%	0.67 [0.22, 2.05]
Kelley 2002	35	274	22	276	9.2%	1.60 [0.97, 2.66]
Krempf 2003	24	346	12	350	6.6%	2.02 [1.03, 3.98]
Lindgärde 2000	10	190	5	186	3.5%	1.96 [0.68, 5.62]
Miles 2002	26	255	13	261	7.1%	2.05 [1.08, 3.89]
Richelsen 2007	8	153	8	156	4.1%	1.02 [0.39, 2.65]
Rössner 2000	15	244	4	243	3.3%	3.73 [1.26, 11.09]
Sjöström 1998	23	345	9	343	5.7%	2.54 [1.19, 5.41]
Swinburn 2005	12	170	8	169	4.7%	1.49 [0.63, 3.56]
Total (95% CI)		**4252**		**3804**	**100.0%**	**1.51 [1.22, 1.89]**
Total events	339		195			

Heterogeneity: Tau² = 0.07; Chi² = 26.95, df = 19 (P = 0.11); I² = 30%
Test for overall effect: Z = 3.71 (P = 0.0002)

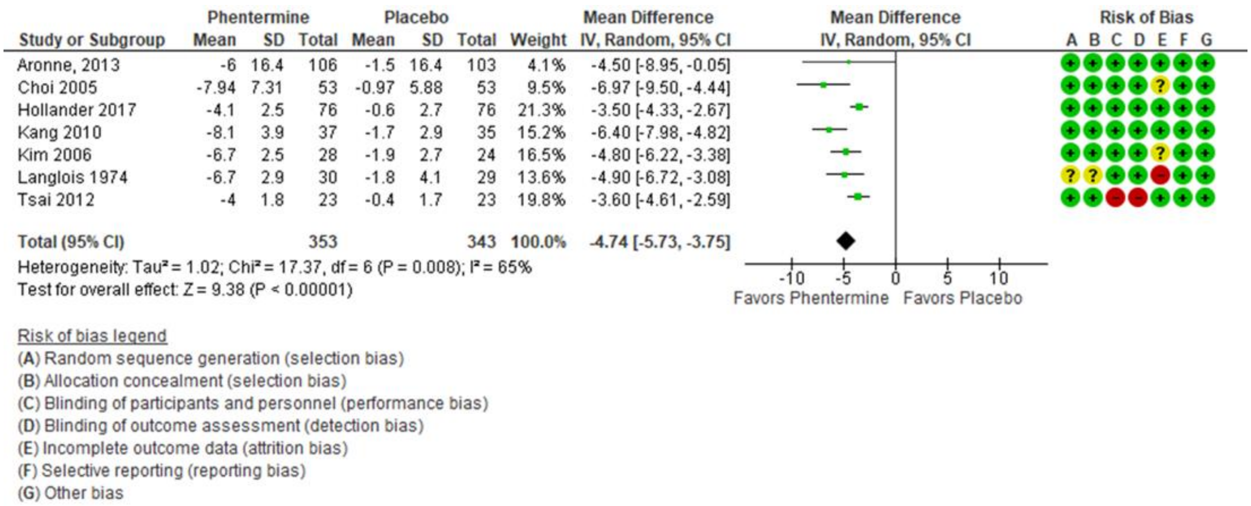


Supplementary Figure 8. Benefits and harms of phentermine over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; **(B)** Mean difference in weight loss (in kg); **(C)** Proportion of patients achieving at least 5% total body weight loss; **(D)** Proportion of patients achieving at least 10% total body weight loss; **(E)** Serious adverse events; **(F)** Discontinuation of treatment due to adverse events

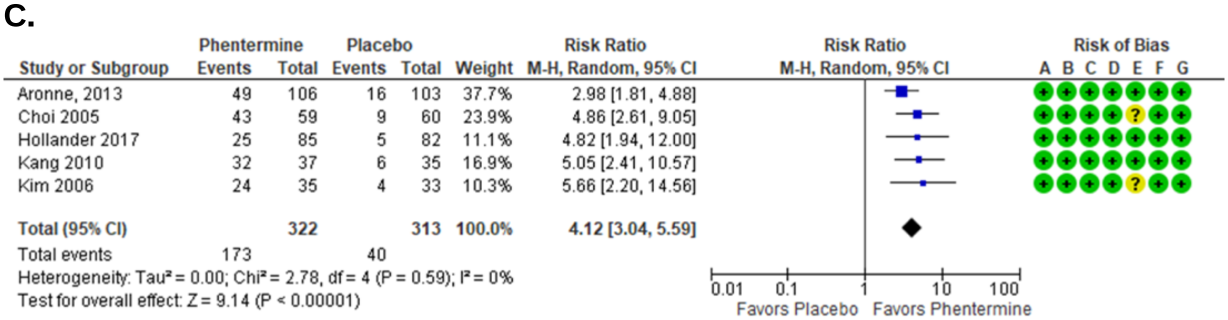
A.



B.

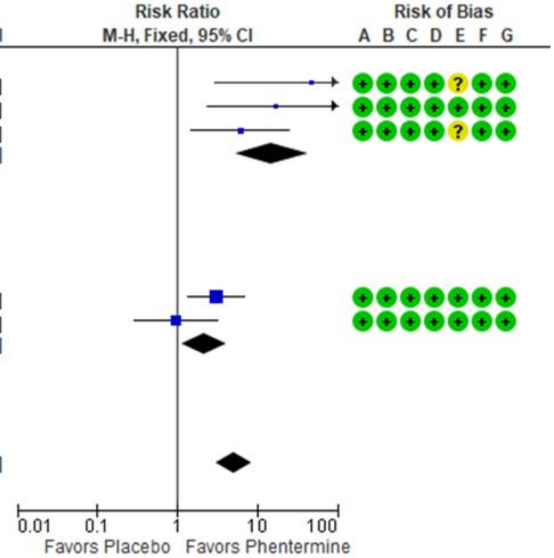


C.



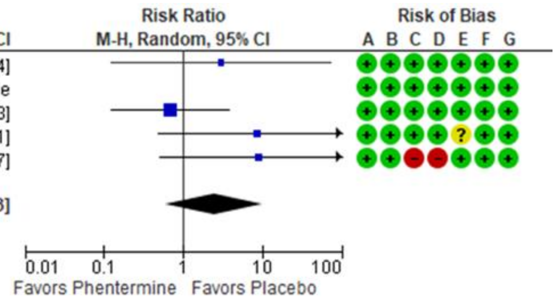
D.

Study or Subgroup	Phentermine Events	Phentermine Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Fixed, 95% CI
1.4.1 Follow up 3 months						
Choi 2005	23	59	0	60	3.1%	47.78 [2.97, 768.97]
Kang 2010	18	37	1	35	6.5%	17.03 [2.40, 120.87]
Kim 2006	13	35	2	33	13.1%	6.13 [1.50, 25.12]
Subtotal (95% CI)		**131**		**128**	**22.7%**	**15.02 [5.27, 42.81]**
Total events	54		3			
Heterogeneity: Chi² = 2.23, df = 2 (P = 0.33); I² = 10%						
Test for overall effect: Z = 5.07 (P < 0.00001)						
1.4.2 Follow up 6 months						
Aronne, 2013	22	106	7	103	45.0%	3.05 [1.36, 6.84]
Hollander 2017	5	85	5	82	32.3%	0.96 [0.29, 3.21]
Subtotal (95% CI)		**191**		**185**	**77.3%**	**2.18 [1.14, 4.17]**
Total events	27		12			
Heterogeneity: Chi² = 2.44, df = 1 (P = 0.12); I² = 59%						
Test for overall effect: Z = 2.36 (P = 0.02)						
Total (95% CI)		**322**		**313**	**100.0%**	**5.10 [3.02, 8.61]**
Total events	81		15			
Heterogeneity: Chi² = 12.93, df = 4 (P = 0.01); I² = 69%						
Test for overall effect: Z = 6.10 (P < 0.00001)						
Test for subgroup differences: Chi² = 9.43, df = 1 (P = 0.002), I² = 89.4%						



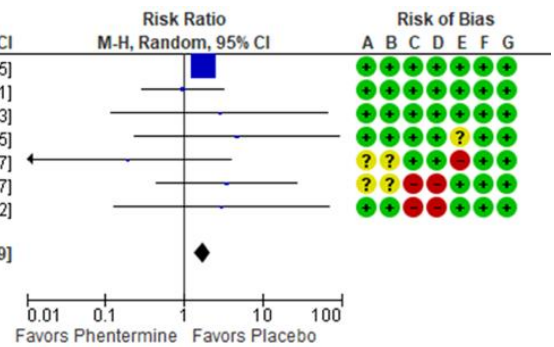
E.

Study or Subgroup	Phentermine Events	Phentermine Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Aronne, 2013	1	109	0	109	16.8%	3.00 [0.12, 72.84]
Hollander 2017	0	85	0	82		Not estimable
Kang 2010	2	28	3	29	43.1%	0.69 [0.12, 3.83]
Kim 2006	4	35	0	33	19.9%	8.50 [0.48, 152.01]
Tsai 2012	4	23	0	23	20.1%	9.00 [0.51, 158.17]
Total (95% CI)		**280**		**276**	**100.0%**	**2.44 [0.60, 10.03]**
Total events	11		3			
Heterogeneity: Tau² = 0.44; Chi² = 3.77, df = 3 (P = 0.29); I² = 20%						
Test for overall effect: Z = 1.24 (P = 0.21)						



F.

Study or Subgroup	Phentermine Events	Phentermine Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Aronne, 2013	241	992	68	492	92.4%	1.76 [1.37, 2.25]
Hollander 2017	5	85	5	82	3.9%	0.96 [0.29, 3.21]
Kang 2010	1	37	0	35	0.6%	2.84 [0.12, 67.53]
Kim 2006	2	35	0	33	0.6%	4.72 [0.24, 94.85]
Langlois 1974	0	30	2	29	0.6%	0.19 [0.01, 3.87]
Munro 1968	7	72	1	36	1.3%	3.50 [0.45, 27.37]
Tsai 2012	1	23	0	23	0.6%	3.00 [0.13, 70.02]
Total (95% CI)		**1274**		**730**	**100.0%**	**1.73 [1.36, 2.19]**
Total events	257		76			
Heterogeneity: Tau² = 0.00; Chi² = 4.07, df = 6 (P = 0.67); I² = 0%						
Test for overall effect: Z = 4.53 (P < 0.00001)						

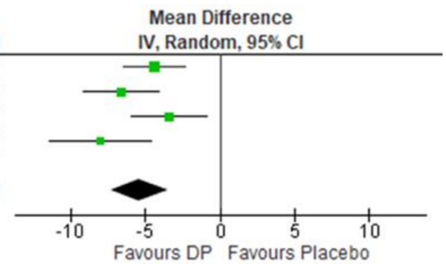


Supplementary Figure 9. Benefits and harms of diethylpropion over placebo in the management of obesity: **(A)** Mean difference in % total body weight loss; **(B)** Mean difference in weight loss (in kg); **(C)** Proportion of patients achieving at least 5% total body weight loss; **(D)** Proportion of patients achieving at least 10% total body weight loss; **(E)** Discontinuation of treatment due to adverse events

A.

	DP			Placebo				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
Abramson 1980	-8.8	4.02	32	-4.4	4.02	28	30.4%	-4.40 [-6.44, -2.36]
Cercato 2009	-9.8	6.9	37	-3.2	3.7	32	25.1%	-6.60 [-9.17, -4.03]
McQuarrie 1975	-5.5	4.04	22	-2.1	4.04	19	25.9%	-3.40 [-5.88, -0.92]
Suplicy 2014	-11.6	7.7	28	-3.6	5	29	18.6%	-8.00 [-11.38, -4.62]
Total (95% CI)			**119**			**108**	**100.0%**	**-5.36 [-7.23, -3.50]**

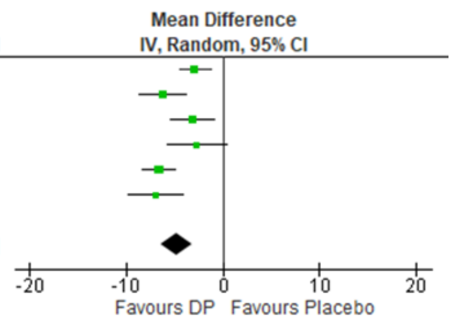
Heterogeneity: Tau² = 1.90; Chi² = 6.38, df = 3 (P = 0.09); I² = 53%
Test for overall effect: Z = 5.63 (P < 0.00001)



B.

	DP			Placebo				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
Abramson 1980	-6.6	3.21	32	-3.7	3.21	28	20.0%	-2.90 [-4.53, -1.27]
Cercato 2009	-9.3	6.6	37	-3.1	3.6	32	16.0%	-6.20 [-8.67, -3.73]
McQuarrie 1975	-4.7	3.73	22	-1.54	3.73	19	16.8%	-3.16 [-5.45, -0.87]
Parsons 1981	-7.2	3.92	12	-4.5	3.92	13	13.4%	-2.70 [-5.78, 0.38]
Soto-Molina 2015	-7.7	4.59	78	-1.1	6.18	78	19.6%	-6.60 [-8.31, -4.89]
Suplicy 2014	-10	6.4	28	-3.1	4.3	29	14.3%	-6.90 [-9.74, -4.06]
Total (95% CI)			**209**			**199**	**100.0%**	**-4.74 [-6.40, -3.08]**

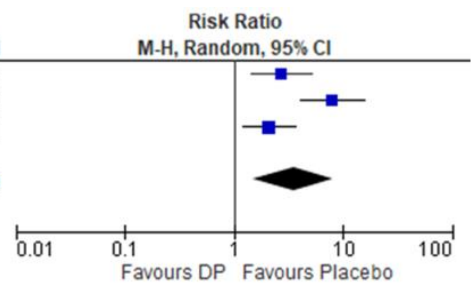
Heterogeneity: Tau² = 2.90; Chi² = 16.54, df = 5 (P = 0.005); I² = 70%
Test for overall effect: Z = 5.60 (P < 0.00001)



C.

	DP		Placebo			Risk Ratio
Study or Subgroup	**Events**	**Total**	**Events**	**Total**	**Weight**	**M-H, Random, 95% CI**
Cercato 2009	25	37	8	32	32.9%	2.70 [1.42, 5.13]
Soto-Molina 2015	64	78	8	78	32.5%	8.00 [4.12, 15.55]
Suplicy 2014	20	28	10	29	34.6%	2.07 [1.19, 3.60]
Total (95% CI)		**143**		**139**	**100.0%**	**3.51 [1.50, 8.18]**
Total events	109		26			

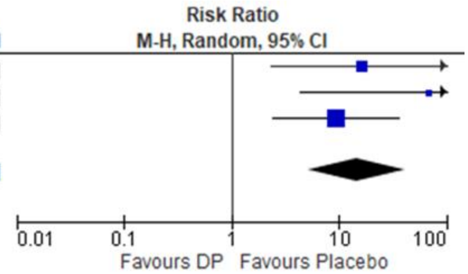
Heterogeneity: Tau² = 0.46; Chi² = 11.27, df = 2 (P = 0.004); I² = 82%
Test for overall effect: Z = 2.90 (P = 0.004)



D.

	DP		Placebo			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Cercato 2009	19	37	1	32	28.2%	16.43 [2.33, 116.01]
Soto-Molina 2015	34	78	0	78	14.0%	69.00 [4.30, 1105.95]
Suplicy 2014	18	28	2	29	57.8%	9.32 [2.38, 36.52]
Total (95% CI)		**143**		**139**	**100.0%**	**14.48 [5.13, 40.87]**
Total events	71		3			

Heterogeneity: Tau² = 0.00; Chi² = 1.99, df = 2 (P = 0.37); I² = 0%
Test for overall effect: Z = 5.05 (P < 0.00001)



E.

	DP		Placebo			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
Abramson 1980	2	32	2	28	26.9%	0.88 [0.13, 5.81]
Cercato 2009	3	37	1	32	19.7%	2.59 [0.28, 23.73]
Parsons 1981	1	12	0	13	10.0%	3.23 [0.14, 72.46]
Soto-Molina 2015	2	78	0	78	10.6%	5.00 [0.24, 102.49]
Suplicy 2014	2	28	3	29	32.9%	0.69 [0.12, 3.83]
Total (95% CI)		**187**		**180**	**100.0%**	**1.37 [0.51, 3.66]**
Total events	10		6			

Heterogeneity: Tau² = 0.00; Chi² = 2.18, df = 4 (P = 0.70); I² = 0%
Test for overall effect: Z = 0.63 (P = 0.53)